

If the coefficients fall off like $1/k^2$, then the Fourier series converges everywhere since

$$|a_k \cos kt + b_k \sin kt| \leq \frac{4M_2}{k^2}$$

The same pattern clearly applies to functions having continuous m th derivatives, except at a finite number of points, the coefficients fall off like $1/k^m$. If it has a continuous m th derivative, the coefficients fall off like $1/k^{m+1}$ (m an integer).

PROBLEMS

Find the Fourier series for:

$$32.3.1 \quad F(t) = \begin{cases} t(\pi - t) & 0 \leq t \leq \pi \\ -f(-t) & 0 \geq t \geq -\pi \end{cases}$$

$$32.3.2 \quad F(t) = 1 - t^2/\pi^2 \quad -\pi \leq t \leq \pi$$

$$32.3.3 \quad F(t) = x(1 - x^2) \quad -1 \leq x \leq 1$$

32.3.4 Discuss the rate of convergence of

$$y(t) = 3t^5 - 10t^3 + 7t \quad -1 \leq t \leq 1$$

32.4 IMPROVING THE CONVERGENCE

Given a function to expand in a Fourier series, we examine the discontinuities in the function and in its derivatives. Suppose, for example, we have the common case of a discontinuity in the function at the ends of the interval. This discontinuity will control the rate of convergence and make it rather slow. If we were to subtract the function (see Fig. 32.4.1)

$$y(x) = a + bx$$

with the value of b picked so that the difference between the given function $F(x)$ and $y(x)$ is continuous at the endpoints, then the convergence of the result would be more rapid.

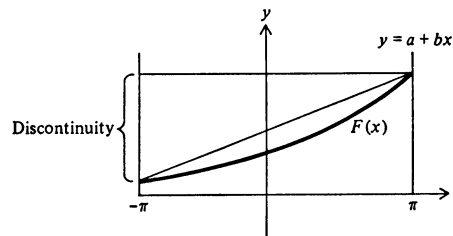


FIGURE 32.4.1

And this is a general approach. We can subtract simple functions, typically polynomials but not necessarily only these, so that the difference has as much smoothness as we wish, and hence the resulting Fourier series will converge as rapidly as we please—thus minimizing the aliasing troubles that occur due to the sampling rate used. For this purpose we need some standard functions with various singularities.

As an example of such a standard function, suppose we have the function

$$y(t) = t/2\pi \quad -\pi < t < \pi$$

which has a discontinuity of unit height at $t = \pi$. It is easy to see that

$$y(t) = -y(-t)$$

so that the $a_k = 0$. The coefficients of the sine terms are

$$\begin{aligned} b_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} y(t) \sin kt \, dt = \frac{2}{\pi} \int_0^{\pi} \frac{t}{2\pi} \sin kt \, dt \\ &= \frac{1}{\pi^2} t \left(-\frac{\cos kt}{k} \right) \Big|_0^{\pi} + \frac{1}{k\pi^2} \int_0^{\pi} \cos kt \, dt \\ &= \frac{\pi}{\pi^2 k} [(-1)\cos \pi k] = \frac{(-1)^{k-1}}{k\pi} \end{aligned}$$

Hence

$$y(t) = \frac{1}{\pi} \left(\sin t - \frac{\sin 2t}{2} + \frac{\sin 3t}{3} - \frac{\sin 4t}{4} + \cdots \right)$$

This series converges to $y(t)$ for all values of t *except* at the discontinuity ($t = \pi$) where clearly it converges to zero (the average of the two values $1/2$ and $-1/2$ which are the limit values as $t \rightarrow \pi$ or $t \rightarrow -\pi$ from the origin). See Fig. 32.4.2.

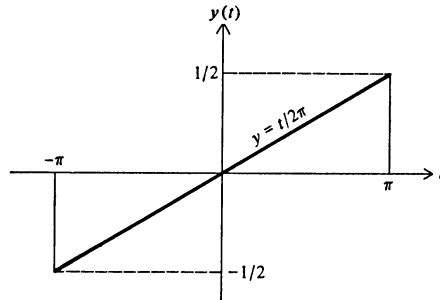


FIGURE 32.4.2

If we set

$$t = t' - a + \pi$$

we translate the discontinuity to $t' = a$. In place of the original series we have (dropping the prime on t) for a function $y(t)$ having a unit jump at $t = a$

$$\begin{aligned} y(t) &= \frac{1}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \sin[(kt - ka) + k\pi] \\ &= \frac{1}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \sin(kt - ka) \cos k\pi = \frac{1}{\pi} \sum_{k=1}^{\infty} \frac{\sin(ka - kt)}{k} \\ &= \frac{1}{\pi} \sum_{k=1}^{\infty} \frac{\sin ka}{k} \cos kt - \frac{1}{\pi} \sum_{k=1}^{\infty} \frac{\cos ka}{k} \sin kt \end{aligned}$$

In this Fourier series the coefficients again fall off as $1/k$.

If we now consider a function with a finite number M of simple discontinuities and with straight lines between the discontinuities, then if the discontinuity at $t = a_i$ is of size y_i , we take the appropriate linear combination of the expansions (including possibly an a_0 term)

$$\begin{aligned} y(t) &= \frac{1}{\pi} \sum_{i=1}^M y_i \left(\sum_{k=1}^{\infty} \frac{\sin ka_i}{k} \cos kt - \sum_{k=1}^{\infty} \frac{\cos ka_i}{k} \sin kt \right) \\ &= \frac{1}{\pi} \sum_{k=1}^{\infty} \left(\sum_{i=1}^M y_i \sin ka_i \right) \frac{\cos kt}{k} - \frac{1}{\pi} \sum_{k=1}^{\infty} \left(\sum_{i=1}^M y_i \cos ka_i \right) \frac{\sin kt}{k} \end{aligned}$$

where we have formally interchanged the summation processes.

Similar expansions can easily be found for functions having other types of discontinuities and need not be given here.

PROBLEMS

32.4.1 Find an expansion for a unit discontinuity in the derivative at $t = a$.

32.4.2 Do the same as in Prob. 32.4.1 except for the second derivative.

32.5 THE GIBBS PHENOMENON¹

In order to understand better what happens at a discontinuity, we begin with the special case of a rectangular wave $H(t)$ of period 2π (see Fig. 32.5.1). If we compute the partial sum of the first $2n$ terms, the cosine terms are all zero, and we have (in the continuous case)

$$H_{2n}(t) = \frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^n \frac{1}{2k-1} \sin(2k-1)t$$

¹ Much of this is taken from a Bell Telephone Laboratories memorandum by R. G. Segers. See also Ref. 33.

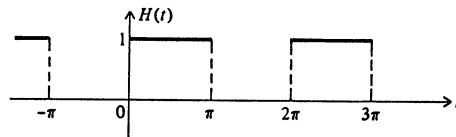
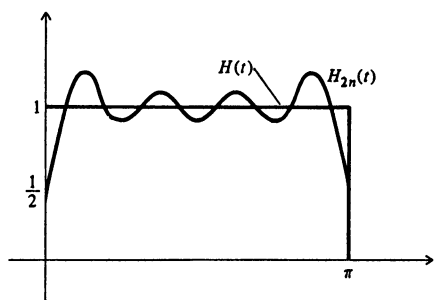


FIGURE 32.5.1

Gibbs (in 1899) pointed out that the partial sum H_{2n} overshoots the function by a certain amount (see Fig. 32.5.2). More precisely

$$H_{2n}\left(\frac{\pi}{2n}\right) \rightarrow 1.08949 \cdots \quad \text{as } n \rightarrow \infty$$

Indeed, not only does $H_{2n}(t)$ overshoot the function $H(t)$, but it also tends to oscillate about $H(t)$, and the oscillations decrease slowly as we move away from the discontinuity.

FIGURE 32.5.2
The Gibbs phenomenon.

In order to explain the phenomenon, we write the expansion as

$$\begin{aligned} H_{2n}(t) &= \frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^n \int_0^t \cos(2k-1)x \, dx \\ &= \frac{1}{2} + \frac{2}{\pi} \int_0^t \sum_{k=1}^n \cos(2k-1)x \, dx \\ &= \frac{1}{2} + \frac{1}{\pi} \int_0^t \frac{\sin 2nx}{\sin x} \, dx \end{aligned}$$

where we have used

$$\sum_{k=1}^n \cos(2k-1)x = \frac{\sin 2nx}{2 \sin x}$$

From this it is clear that the maxima and minima occur (for $0 \leq t \leq \pi$) at the points where the derivative of $H_{2n}(t) = 0$, namely,

$$\frac{dH_{2n}(t)}{dt} = \frac{1}{\pi} \frac{\sin 2nt}{\sin t} = 0$$

which gives

$$t = \frac{m\pi}{2n} \quad m = 1, 2, \dots, 2n-1$$

and that they alternate. The size of these has been computed by Carslaw [7].

What is true for this special function is clearly true for more general functions since the discontinuity can be regarded as coming from a rectangular wave *added* to a smooth function.

32.6 LANCZOS' σ FACTORS

As our second approach to improving the convergence, we propose to replace the rapidly oscillating function $H_{2n}(t)$ by a smoothed function

$$\overline{H}_{2n}(t) = \frac{n}{\pi} \int_{t-\pi/2n}^{t+\pi/2n} H_{2n}(s) ds$$

where we have averaged over one complete oscillation of $H_{2n}(t)$ centered about t . We now find the effect of the smoothing.

Using the earlier expansion of the function $H_{2n}(t)$, we get

$$\begin{aligned} \overline{H}_{2n}(t) &= \frac{n}{\pi} \int_{t-\pi/2n}^{t+\pi/2n} \left[\frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^n \frac{1}{2k-1} \sin(2k-1)s \right] ds \\ &= \frac{n}{\pi} \left[\frac{\pi}{2n} + \frac{2}{\pi} \sum_{k=1}^n \frac{-1}{(2k-1)^2} \cos(2k-1)s \right]_{t-\pi/2n}^{t+\pi/2n} \\ &= \frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^n \frac{1}{2k-1} \frac{\sin[(2k-1)(\pi/2n)]}{(2k-1)(\pi/2n)} \sin(2k-1)t \end{aligned}$$

If we compare this with $H_{2n}(t)$, we find that *we now have the extra factor in the expansion*

$$\sigma_{2k-1} = \frac{\sin(2k-1)(\pi/2n)}{(2k-1)(\pi/2n)}$$

for each term in the summation.

The effect of this factor σ_k is to reduce the limiting value of the maximum overshoot from 0.08949 to 0.01187 and the first minimum from 0.04859 to 0.00473, etc. Thus the Gibbs phenomenon has been greatly reduced by putting in the σ factors which arose from the smoothing of $H_{2n}(t)$ over a short interval of length π/n .

32.7 THE σ FACTORS IN THE GENERAL CASE

We have studied the special case of a rectangular wave; we now show that the weight factors σ_k are the same for any Fourier series.

Let $f(t)$ ($0 \leq t \leq 2\pi$) be integrable, and let

$$a_k = \frac{1}{\pi} \int_0^{2\pi} f(t) \cos kt \, dt \quad k = 0, 1, 2, \dots$$

$$b_k = \frac{1}{\pi} \int_0^{2\pi} f(t) \sin kt \, dt \quad k = 1, 2, \dots$$

be the Fourier coefficients¹

$$f_n(t) = \frac{a_0}{2} + \sum_{k=1}^{n-1} (a_k \cos kt + b_k \sin kt) + \frac{a_n}{2} \cos nt$$

In the previous special case we had frequencies up to $2n$; this time we have chosen those up to n only, and so averaging over the interval $(t - \pi/n, t + \pi/n)$, we get

$$\begin{aligned} \hat{f}_n(t) &= \frac{1}{2\pi/n} \int_{t-\pi/n}^{t+\pi/n} f_n(s) \, ds \\ &= \frac{n}{2\pi} \left[\frac{a_0}{2} \frac{2\pi}{n} + \sum_{k=1}^{n-1} \left(a_k \frac{\sin ks}{k} - b_k \frac{\cos ks}{k} \right) \right]_{t-\pi/n}^{t+\pi/n} + \frac{a_n}{2} \frac{\sin ns}{n} \Big|_{t-\pi/n}^{t+\pi/n} \\ &= \frac{a_0}{2} + \frac{n}{2\pi} \sum_{k=1}^{n-1} \left\{ \frac{a_k}{k} \left[\sin k \left(t + \frac{\pi}{n} \right) - \sin k \left(t - \frac{\pi}{n} \right) \right] \right. \\ &\quad \left. - \frac{b_k}{k} \left[\cos k \left(t + \frac{\pi}{n} \right) - \cos k \left(t - \frac{\pi}{n} \right) \right] \right\} \\ &\quad + \frac{a_n}{2n} \left[\sin n \left(t + \frac{\pi}{n} \right) - \sin n \left(t - \frac{\pi}{n} \right) \right] \\ &= \frac{a_0}{2} + \frac{n}{2\pi} \sum_{k=1}^{n-1} \left(\frac{a_k}{k} 2 \cos kt \sin \frac{k\pi}{n} + \frac{b_k}{k} \cdot 2 \sin kt \sin \frac{k\pi}{n} \right) \\ &\quad + \frac{a_n}{2n} 2 \cos nt \sin \frac{n\pi}{n} \\ &= \frac{a_0}{2} + \sum_{k=1}^{n-1} \frac{\sin(k\pi/n)}{k\pi/n} (a_k \cos kt + b_k \sin kt) \end{aligned}$$

¹ If we use the conventional form without $1/2$ the last term, the result is the same.

the last term a_n having a zero coefficient. Using the same σ factors (notice we now are using n in place of $2n$)

$$\sigma_k(n) = \frac{\sin(k\pi/n)}{k\pi/n}$$

we get, as before, the smoothed values

$$\hat{f}_n(t) = \frac{a_0}{2} + \sum_{k=1}^{n-1} \sigma_k(n) (a_k \cos kt + b_k \sin kt)$$

Notice that $\sigma_k(n)$ is a function of n and that

$$\sigma_n(n) = \frac{\sin(n\pi/n)}{n\pi/n} = 0$$

Thus to smooth the series, we merely apply the $\sigma_k(n)$ factors (which taper from $\sigma_0 = 1$ to $\sigma_n = 0$) to both the sine and cosine terms. The effect of the σ factors is to greatly smooth the truncated Fourier series.

32.8 A COMPARISON OF CONVERGENCE METHODS

Besides the σ -factor method of Lanczos there is the well-known Fejer method of using the arithmetic means of the partial sums. Fejer's method completely eliminates the oscillation, while Lanczos' method greatly dampens it. As a result, Fejer's method pays a large price in producing, for finite n , a slow-rise time for the partial sum near the discontinuity. Fig. 32.8.1 shows the appropriate comparison of the

Fourier series
Fejer sum
Lanczos σ -factor curves

for a 12-term approximation to a rectangular wave. The advantages of the σ -factor method are clearly apparent.

Fig. 32.8.2 shows the rapidity with which the σ -factor method approaches the rectangular wave as a function of n . Thus even for moderate n the "rise time" of the curve is very short.

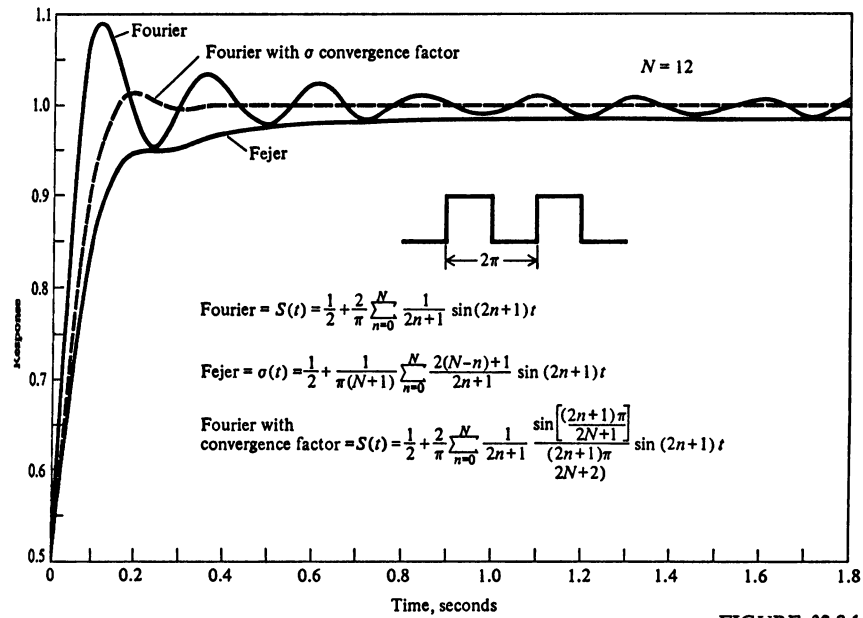


FIGURE 32.8.1

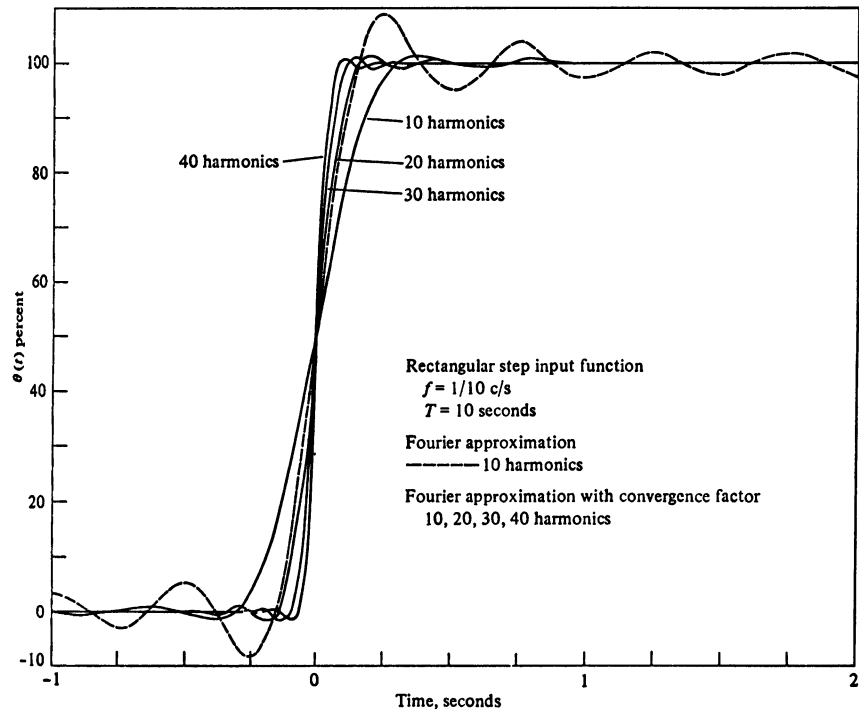


FIGURE 32.8.2

32.9 LANCZOS' DIFFERENTIATION TECHNIQUE

It is sometimes necessary to differentiate a Fourier series. But if we do so, we usually get what is not wanted in the physical problem, namely, a large high-frequency oscillation.

Lanczos [33] introduced a practical approach, namely, that when we differentiate a truncated Fourier-series approximation, we use

$$\frac{f(t + \pi/n) - f(t - \pi/n)}{2\pi/n}$$

as an estimate of the derivative of $f(t)$ at the point t (instead of the usual limit process involved in taking the derivative which has dubious physical significance). Note that the same n appears both in the formula and in the order of the truncated series

$$\begin{aligned} f(t) &= \frac{a_0}{2} + \sum_{k=1}^n (a_k \cos kt + b_k \sin kt) \\ \frac{f(t + \pi/n) - f(t - \pi/n)}{2\pi/n} &= \sum_{k=1}^n \left\{ a_k \left[\cos k \left(\frac{t + \pi}{n} \right) - \cos k \left(\frac{t - \pi}{n} \right) \right] \right. \\ &\quad \left. + b_k \left[\sin k \left(\frac{t + \pi}{n} \right) - \sin k \left(\frac{t - \pi}{n} \right) \right] \right\} \frac{n}{2\pi} \\ &= \sum (-ka_k \sin kt + kb_k \cos kt) \sigma_k \end{aligned}$$

Thus the same σ factors as before, put in the formally differentiated series, give the appropriate result.

32.10 SUMMARY

We have given two methods for improving the convergence of a Fourier series. The method of singularities subtracts out the troubles that cause the slow convergence before the series is evaluated. Thus the numerical computation on the difference has a rapidly converging Fourier series, and the aliasing that arises from the sampling does not cause serious trouble. The analytical results from the analytical expansions supply the other coefficients and keep the aliasing from falling into the lower frequencies.

The second method of σ_k factors applies to series *after* they are known and damps down the Gibbs phenomena at discontinuities. It is chiefly applicable to analytical expansions, such as those we derived above and used to subtract out the singularities.

THE FAST FOURIER TRANSFORM

33.1 THE DIRECT CALCULATION

The Fourier coefficients of a function $F(t)$, and more often the power spectra, are used in many applications. The coefficients may be found directly from the formulas

$$a_k = \frac{1}{\pi} \int_0^{2\pi} F(t) \cos kt \, dt \quad k = 0, 1, 2, \dots$$

$$b_k = \frac{1}{\pi} \int_0^{2\pi} F(t) \sin kt \, dt \quad k = 1, 2, 3, \dots$$

From Sec. 31.8 we see that the equally spaced, equally weighted samples provide an excellent way of evaluating these integrals in case they cannot be done in closed form. They are effectively the Gaussian quadrature formulas and have, therefore, the corresponding accuracy associated with such formulas.

Another approach to finding the power spectrum of a function is to take the equally spaced samples (with the consequent aliasing) and fit a finite Fourier series to them. A little thought shows that these two, apparently alternate, approaches are equivalent.

The problem of aliasing appears more complicated than it is in reality. Suppose, for example, that the function has a discontinuity at the ends of the interval. This could be removed by subtracting a straight line. If then we do the sampling, there will be less aliasing due to the sampling than before because the series now converges more rapidly.

On the other hand, suppose we sample first and then from the discrete samples remove the (same) straight line. The numbers we operate upon are the same in the two cases. How do we reconcile this with the claim that once the aliasing occurs it cannot be undone? The answer lies in understanding properly "the aliasing cannot be undone." Clearly, if we have a frequency above the folding frequency and then sample, it will appear as if it were in the band, and we cannot *know* that it is not the frequency inside the band. On the other hand, we may at times do operations that cause further aliasing, and it can happen that sometimes these operations undo an earlier aliasing—it is simply that without further information there are aliases that we cannot *know* will be undone.

33.2 INTRODUCTION TO THE FAST FOURIER TRANSFORM (FFT)

The fast Fourier transform was rediscovered and adequately publicized by Cooley and Tukey,¹ though it had been discovered a number of times before, and was to some extent understood in the general literature. The explicit recognition of its importance is one of the few significant advances in numerical analysis in the past few decades. Much has been learned about various aspects of computation, but no other recent discovery has so profoundly affected so many different fields as has the fast Fourier transform.

For N samples the direct calculation, as indicated in the previous section, would involve approximately N^2 multiplications and N^2 additions. The fast Fourier transform has, under suitable circumstances, reduced the number of operations to something like $N \log N$. For N small, say around a hundred or so, the change is not tremendous, but when N is around 10,000 as it is in many applications, the effect is to reduce the computing to below 1 percent of the direct method and to make what were totally impractical computations now a matter of routine processing.

The fast Fourier transform is still as of 1971 in a state of rapid evolution with numerous papers appearing each year. Thus the treatment we give must be limited to the basics, and cannot be definitive. Before committing oneself to a

¹ J. W. Cooley and J. W. Tukey, An Algorithm for the Machine Calculation of Complex Fourier Series, *Math. of Comp.*, vol. 19, pp. 297–301, April, 1965.

large project using the fast Fourier transform, an examination of the current literature should be made (probably the *IEEE Transactions on Electronic Computers* should be examined first).

33.3 THE CENTRAL IDEA OF THE FAST FOURIER TRANSFORM

The fast Fourier transform is based on a method of factoring the transform into the product of two transforms—a product in the sense that one transform *follows* the other. As a result of the factoring, if we have M data points and if M can be factored into

$$M = GH$$

then in place of M^2 multiplications and additions we get approximately

$$M(G + H)$$

operations of each type. Repeated applications of the factoring leads to the following result. If

$$M = m_1 m_2 \cdots m_k$$

then we will have approximately

$$M(m_1 + m_2 + \cdots + m_k)$$

operations. In the most favorable case, when M is a power of 2, say 2^k , we have

$$M(2k)$$

operations, where $k = \log_2 M$. Thus we have approximately $M \log M$ operations in place of M^2 operations. In other cases, where M has many small factors, somewhat the same effect of greatly decreasing the number of operations happens.

It is easiest to demonstrate the way the factorization occurs in terms of complex notation, where $M = GH$ and the coefficient is given by

$$\begin{aligned} C_k \equiv C(k) &= \frac{1}{GH} \sum_{p=0}^{M-1} F(x_p) e^{-2\pi i k x_p} \\ &= \frac{1}{GH} \sum_{p=0}^{GH-1} F\left(\frac{Lp}{M}\right) e^{-2\pi i k Lp/M} \end{aligned}$$

using $x_p = Lp/M$ ($p = 0, 1, \dots, M-1$). For convenience we set $L = 1$.

The heart of the method is writing the two indices k and p in a suitable way. Divide k by G to get

$$k = k_1 G + k_0$$

Similarly divide p by H to get

$$p = p_1 H + p_0$$

where, of course, $k_0 < G$, $k_1 < H$, $p_0 < H$, and $p_1 < G$. With this notation the formula becomes (the double sum covers the *same numbers* as the original single sum)

$$\begin{aligned} C(k_1 G + k_0) &= \frac{1}{GH} \sum_{p_0=0}^{H-1} \sum_{p_1=0}^{G-1} F\left(\frac{p_1 H + p_0}{GH}\right) e^{-2\pi i(k_1 G + k_0)(p_1 H + p_0)/(GH)} \\ &= \frac{1}{GH} \sum_{p_0=0}^{H-1} e^{-2\pi i k_0 p_0 / GH} e^{-2\pi i k_1 p_0 / H} \left[\sum_{p_1=0}^{G-1} F\left(\frac{p_1}{G} + \frac{p_0}{GH}\right) e^{-2\pi i k_0 p_1 / G} \right] \end{aligned}$$

where we have used the fact that

$$e^{-2\pi i k_1 p_1} \equiv 1$$

and suppressed those terms.

The square brackets effectively contain the Fourier coefficient of the expansion of $1/H$ of the samples, phase shifted $p_0/(GH)$. There are H such sums to be done, one for each p_0 . Let us label these sums

$$\hat{C}(k_0, p_0) = \sum_{p_1=0}^{G-1} F\left(\frac{p_1}{G} + \frac{p_0}{GH}\right) e^{-2\pi i k_0 p_1 / G}$$

and we have

$$C(k_1 G + k_0) = \frac{1}{GH} \sum_{p_0=0}^{H-1} \hat{C}(k_0, p_0) e^{-2\pi i[k_0/(GH) + k_1/H]p_0}$$

which are the coefficients of a second Fourier expansion, this time of the $\hat{C}(k_0, p_0)$, and there are H terms to be evaluated in the sum.

A count of the operations shows that they are proportional to

$$GH(G + H)$$

as claimed.

33.4 THE FAST FOURIER TRANSFORM IN PRACTICE

At first it was believed that using the factors of 2 as often as possible was the best way of implementing the fast Fourier transform. Then it was realized that when there was a pair of factors of 2, it was best handled as a factor of 4, because of the fact that the values of the trigonometric functions were all 0s and 1s. Thus it was advocated that the G (or H) be taken as 4 whenever possible, and if there was an odd number of 2s, then only one stage should use a factor of 2. Clearly the factor of 4 uses only additions and subtractions.

More recently a radix-8 fast Fourier transform has been recommended. However, the situation is still in a state of flux and the final word is not in. For further references see the sources cited below.¹

Because of the importance of taking Fourier transforms, the fast Fourier transform has been implemented in hardware directly in a number of cases, and is reasonably available. The hardware implementation makes further spectacular improvements in speed (and cost) of the computation.

33.5 DANGERS OF THE FOURIER TRANSFORM

The fast Fourier transform calculates (within roundoff) the same quantities as does the direct method, so that the troubles we shall discuss are not functions of the fast Fourier transform but rather of the Fourier transform itself.

First, there is the ever present problem of *aliasing* (Secs. 31.2 and 31.6).

Second, there is the problem of *leakage* due to the use of a finite length of record. The effect on a single spike is that it is replaced by a $(\sin x)/x$ ripple, and on a continuous spectrum, a large peak tends to "leak out" into the nearby parts of the spectrum (see Sec. 34.8).

Finally, there is the *picket-fence effect*. Each value of the sample spectrum acts like a spike, and the sum of the $(\sin x)/x$ terms, each shifted to its proper place, gives the spectrum the appearance of a sort of picket-fence effect of equally spaced ripples. This is noticeable if other than the properly spaced frequencies are computed.

These effects will not be discussed in the text at greater length because of the theoretical difficulties involved. We mention them here only as a matter of caution.

33.6 FOURIER ANALYSIS USING 12 POINTS

Because the fast Fourier transform is often so obscure when first seen, we give the classic, closely related 12-point case for the Fourier series. In this example the reduction is made to follow from the periodicity of the trigonometric functions, which is in fact what the fast Fourier transform actually uses. But since it is in the real domain, it may be easier for some to follow—for others because of the clumsy symbolism it may be harder!

¹ G. D. Bergland, A Guided Tour of the Fast Fourier Transform, *IEEE Spectrum*, July, 1969, pp. 41–52. W. M. Gentleman and G. Sande, Fast Fourier Transforms—For Fun and Profit, *Proc. FJCC*, 1966, pp. 563–578.

The special case of 12 points ($N = 6$) is of interest because it occurs frequently and is easy to do, even by hand! We have

$$6a_k = \sum_{x=0}^{11} f(x) \cos \frac{\pi}{6} kx$$

$$6b_k = \sum_{x=0}^{11} f(x) \sin \frac{\pi}{6} kx$$

If we break up each sum into the ranges 0 to 6 and 7 to 11 in x , and in the second range put $x = 12 - x'$, we get

$$6a_k = \sum_{x=0}^6 f(x) \cos \frac{\pi}{6} kx + \sum_{x'=1}^5 f(12 - x') \cos \frac{\pi}{6} kx'$$

$$6b_k = \sum_{x=0}^6 f(x) \sin \frac{\pi}{6} kx - \sum_{x'=1}^5 f(12 - x') \sin \frac{\pi}{6} kx'$$

This suggests writing

	$f(0)$	$f(1)$	$f(2)$	$f(3)$	$f(4)$	$f(5)$	$f(6)$
		$f(11)$	$f(10)$	$f(9)$	$f(8)$	$f(7)$	
Add	$s(0)$	$s(1)$	$s(2)$	$s(3)$	$s(4)$	$s(5)$	$s(6)$
Subtract		$t(1)$	$t(2)$	$t(3)$	$t(4)$	$t(5)$	

$$6a_k = \sum_{x=0}^6 s(x) \cos \frac{\pi}{6} kx$$

$$6b_k = \sum_{x=1}^5 t(x) \sin \frac{\pi}{6} kx$$

We again break up the range into two parts, this time 0 to 3 and 4 to 6, setting $x = 6 - x'$ in the second range.

$$6a_k = \sum_{x=0}^3 s(x) \cos \frac{\pi}{6} kx + (-1)^k \sum_{x'=0}^2 s(6 - x') \cos \frac{\pi}{6} kx'$$

$$6b_k = \sum_{x=1}^3 t(x) \sin \frac{\pi}{6} kx - (-1)^k \sum_{x'=1}^2 t(6 - x') \sin \frac{\pi}{6} kx'$$

This suggests writing

	$s(0)$	$s(1)$	$s(2)$	$s(3)$	$t(1)$	$t(2)$	$t(3)$
	$s(6)$	$s(5)$	$s(4)$		$t(5)$	$t(4)$	
Add	$u(0)$	$u(1)$	$u(2)$	$u(3)$	$p(1)$	$p(2)$	$p(3)$
Subtract	$v(0)$	$v(1)$	$v(2)$		$q(1)$	$q(2)$	

The result, when written out, is

$$6a_0 = u(0) + u(1) + u(2) + u(3) = [u(0) + u(3)] + [u(1) + u(2)]$$

$$6a_1 = v(0) + \frac{\sqrt{3}}{2}v(1) + \frac{1}{2}v(2) = \left[v(0) + \frac{1}{2}v(2)\right] + \frac{\sqrt{3}}{2}v(1)$$

$$6a_2 = u(0) + \frac{1}{2}[u(1) - u(2)] - u(3) = [u(0) - u(3)] + \frac{1}{2}[u(1) - u(2)]$$

$$6a_3 = v(0) - v(2)$$

$$6a_4 = u(0) - \frac{1}{2}u(1) - \frac{1}{2}u(2) + u(3) = [u(0) + u(3)] - \frac{1}{2}[u(1) + u(2)]$$

$$6a_5 = v(0) - \frac{\sqrt{3}}{2}v(1) + \frac{1}{2}v(2) = \left[v(0) + \frac{1}{2}v(2)\right] - \frac{\sqrt{3}}{2}v(1)$$

$$6a_6 = u(0) - [u(1) - u(2)] - u(3) = [u(0) - u(3)] - [u(1) - u(2)]$$

$$6b_1 = \frac{1}{2}p(1) + \frac{\sqrt{3}}{2}p(2) + p(3) = \left[\frac{1}{2}p(1) + p(3)\right] + \frac{\sqrt{3}}{2}p(2)$$

$$6b_2 = \frac{\sqrt{3}}{2}[q(1) + q(2)]$$

$$6b_3 = p(1) - p(3)$$

$$6b_4 = \frac{\sqrt{3}}{2}[q(1) - q(2)]$$

$$6b_6 = \frac{1}{2}p(1) - \frac{\sqrt{3}}{2}p(2) + p(3) = \left[\frac{1}{2}p(1) + p(3)\right] - \frac{\sqrt{3}}{2}p(2)$$

In the process we do less than 60 arithmetic operations, and most of them are simple additions. This compares favorably with the $6N^2 = 6^3 = 216$ needed by the direct method of Sec. 33.1. The programming, of course, is a bit longer to write, but is very easy since there is no complex logic and since only one difficult number $\sqrt{3}/2 = 0.8660254075$ is required, and this only four times. It is remarkably easy to do by hand.

These general methods can be, and have been, applied in cases other than $N = 6$.

The central idea in both Secs. 33.3 and 33.6 is to cause the additions to occur *before* the multiplications whenever possible.

PROBLEMS

33.6.1 If $f(x) = x(12 - x)$ for $x = 0, 1, \dots, 11$, find, by hand, the Fourier expansion of $f(x)$.

33.6.2 Do the same for $f(x) = \begin{cases} 0 & 0 \leq x < 6 \\ 1 & 7 \leq x < 12 \end{cases}$

33.6.3 Carry out an analysis using the midpoints t_p .

33.7 COSINE EXPANSIONS

If we assume that the function $f(x)$ is periodic of period $2N$, then whether we use $x = 0, 1, \dots, 2N-1$ or the sequence $-(N-1), -(N-2), \dots, 0, 1, \dots, N$, the result is the same. Now suppose that we have a function defined for $x = 0, 1, \dots, N$ and we wish to expand it in a series of cosine terms alone. Let us *define* $f(-x) \equiv f(x)$ for $x = 1, 2, \dots, N-1$. The function $f(x)$ is now an *even function*. When we make the Fourier expansion, we find that all the $b_k = 0$, since

$$b_k = \frac{1}{N} \sum_{x=-N+1}^N f(x) \sin \frac{\pi k}{N} x$$

and the sine is an odd function. Thus we have a cosine expansion. The formulas for the a_k can be simplified:

$$\begin{aligned} a_k &= \frac{1}{N} \sum_{x=-N+1}^{N-1} f(x) \cos \frac{\pi k}{N} x \\ &= \frac{2}{N} \left[\frac{1}{2} f_0 + \sum_{x=1}^{N-1} f(x) \cos \frac{\pi k}{N} x + \frac{1}{2} f_N (-1)^k \right] \end{aligned}$$

By defining $f(x)$ to be an odd function, we can get a sine expansion in a similar fashion.

We also note that if we have a range $0 \leq x \leq R$, the substitution

$$x = \frac{R}{2} (1 + \cos t)$$

makes the function periodic in t as well as an even function of t . This substitution has a number of advantages in matters of speed of convergence.

PROBLEMS

- 33.7.1 If $f(x)$ is defined for the interval $0 \leq x \leq \pi/2$, show how the function may be extended to $-\pi < x \leq \pi$ so that only odd cosine harmonics occur; only even cosine harmonics.
- 33.7.2 Discuss in detail the problem of finding a Fourier series in sines only; in odd (or even) sine harmonics.

33.8 LOCAL FOURIER SERIES

In many problems the phenomena change slowly, and the idea of a "local Fourier series" whose coefficients change slowly with time is quite natural. It is also natural to center the range about the origin and use the values from $-N+1$ to N

rather than from 0 to $2N - 1$. Thus expanding about the value t , we have the coefficients

$$a_m(t) = \frac{1}{N} \sum_{x=-N+1}^N f(x+t) \cos \frac{\pi}{N} mx \quad m = 0, 1, \dots, N$$

$$b_m(t) = \frac{1}{N} \sum_{x=-N+1}^N f(x+t) \sin \frac{\pi}{N} mx \quad m = 1, 2, \dots, N-1$$

It is easy to obtain $a(t+1)$ and $b(t+1)$ from $a(t)$ and $b(t)$. For example,

$$a_m(t+1) = \frac{1}{N} \sum_{x=-N+1}^N f(x+t+1) \cos \frac{\pi}{N} mx$$

Set $1+x = x'$, or $x = x' - 1$, and

$$\begin{aligned} a_m(t+1) &= \frac{1}{N} \sum_{x'=-N+2}^{N+1} f(x'+t) \cos \frac{\pi}{N} m(x'-1) \\ &= \frac{1}{N} \left[\sum_{x'=-N+1}^N f(x'+t) \cos \frac{\pi}{N} m(x'-1) + f(N+1+t) \cos \frac{\pi}{N} m(N) \right. \\ &\quad \left. - f(-N+1+t) \cos \frac{\pi}{N} m(-N) \right] \\ &= \frac{1}{N} \left\{ \sum_{x'=-N+1}^N f(x'+t) \cos \frac{\pi}{N} m(x'-1) \right. \\ &\quad \left. + (-1)^m [f(N+1+t) - f(-N+1+t)] \right\} \end{aligned}$$

Expanding the cosine term, we get

$$\begin{aligned} a_m(t+1) &= a_m(t) \cos \frac{\pi}{N} m + b_m(t) \sin \frac{\pi}{N} m \\ &\quad + \frac{(-1)^m}{N} [f(N-1+t) - f(-N+1+t)] \end{aligned}$$

Note that we use fixed (for a given frequency) multipliers $\cos \pi m/N$ and $\sin \pi m/N$ and add or subtract the two new values in the range.

Similarly for the sine term,

$$b_m(t+1) = -a_m(t) \sin \frac{\pi}{N} m + b(t) \cos \frac{\pi}{N} m$$

PROBLEMS

33.8.1 Derive the formula for $b_m(t+1)$.

34

THE FOURIER INTEGRAL : NONPERIODIC FUNCTIONS

34.1 OUTLINE AND PURPOSE OF CHAPTER

In Chaps. 31 to 33 we approximated periodic functions by linear combinations of periodic functions—sines and cosines. Periodic functions are fairly rare in practice, and we now turn to the study of more general functions, functions which are not necessarily periodic. A very simple example of a nonperiodic function which is the sum of two periodic functions is

$$y(t) = \cos t + \cos(\sqrt{2}t)$$

Since 1 and $\sqrt{2}$ are not commensurate, this function cannot repeat itself exactly.

The idea that we can approximate a general (nonperiodic) function by a sum of periodic functions (sines and cosines) is no more unreasonable than the Taylor-series representation of a periodic function by means of nonperiodic terms, namely $1, x, x^2, \dots$

In the Fourier series we used an infinite but countable number of discrete frequencies to represent a periodic function. In the Fourier integral we shall use

a noncountable number of frequencies to represent the function. In the formal definition of the Fourier-integral representation of $f(t)$

$$f(t) = \int_{-\infty}^{\infty} F(\sigma) e^{2\pi i \sigma t} d\sigma$$

we have frequencies σ corresponding to *all* possible real numbers (not just integers). In practice we will often find that the frequencies are confined to an interval and that $F(\sigma) = 0$ outside of the interval. Such a function will be called *band-limited*. As an example of a band-limited function, consider the usual hi-fi system which can transmit all the frequencies from some fairly low frequency to some upper limit around 20,000 Hz (cycles per second).¹

The purpose of this chapter is to introduce the Fourier integral and a number of associated results that will enable us to answer questions like:

- 1 What is the effect of sampling at equally spaced points?
- 2 How can we reconstruct a band-limited function from its samples (the so-called sampling theorem) so that we can interpolate for the missing values and use it in the analytic substitution method?
- 3 What is the effect of taking a short run of samples from an infinitely long function?

These three questions are typical of those that cannot be intelligently discussed within the framework of polynomial approximation, and they serve to point out some of the advantages of the frequency approach when trying to understand what is happening in an experiment or a computation.

34.2 NOTATION

This section and the next are devoted to introducing the notation we shall use. They do not contain proofs of most of the statements made, and intuitive, but non-rigorous proofs will be supplied in later sections.²

The continuous Fourier series in the interval $-N < t \leq N$ can be written (Sec. 31.3 in a slightly changed notation)

$$f(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos \frac{\pi}{N} kt + \sum_{k=1}^{\infty} b_k \sin \frac{\pi}{N} kt$$

¹ It has been recently decided to use the name of a scientist in place of "cycles per second," in analogy with amperes being "coulombs per second," and Hertz was so honored. Some of us consider it unfortunate that they did not choose Steinmetz instead of Hertz so that we could use his initials (Charles Proteus Steinmetz).

² We again refer the reader to Lighthill [37], and Lanczos' books [33–35] for more details; we cannot supply a course in the Fourier integral in one short chapter.

where

$$\begin{aligned} a_k &= \frac{1}{N} \int_{-N}^N f(t) \cos \frac{\pi}{N} t \, dt & k = 0, 1, 2, \dots \\ b_k &= \frac{1}{N} \int_{-N}^N f(t) \sin \frac{\pi}{N} t \, dt & k = 1, 2, \dots \end{aligned}$$

or it can be written in the complex notation (Sec. 31.4)

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{(\pi i/N)kt}$$

where

$$c_k = \frac{1}{2N} \int_{-N}^N f(t') e^{-(\pi i/N)kt'} \, dt'$$

and

$$c_k = \begin{cases} \frac{a_k - ib_k}{2} & k > 0 \\ a_0 & k = 0 \\ \frac{a_k + ib_k}{2} & k < 0 \end{cases}$$

The complex form of the Fourier series is, as we have already seen, much easier to handle in theoretical discussions, but of course we still compute with real numbers, and the complex notation is apt to conceal a great deal of computing. Since our use of the Fourier integral is mainly for understanding what is going on in a computation and for formal derivations of results, we shall use the complex form rather than the equivalent real form.

In the complex form both positive and negative frequencies occur, and this is apt to bother the beginner. The two arise formally from the fact that in the real case we have two functions at each positive frequency, the sine and the cosine, and we need two functions to maintain the linear independence at each frequency.

34.3 SUMMARY OF RESULTS

This section summarizes the results of the rest of the chapter and in general does not try to prove things. First, we shall show that corresponding to the Fourier-series representation of a periodic function there is the Fourier-integral representation of a general function

$$f(t) = \int_{-\infty}^{\infty} F(\sigma) e^{2\pi i \sigma t} \, d\sigma$$

where

$$F(\sigma) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i \sigma t} dt$$

The quantity $F(\sigma) d\sigma$ corresponds, loosely, to the c_k in the Fourier series.

$F(\sigma)$ is the *density function* which describes how much a frequency σ is present in the function $f(t)$. The density times the interval $\Delta\sigma$ gives the amount in measurable units. The function $F(\sigma)$ is called the *transform* of $f(t)$. It is customary to use capital and lowercase letters to represent the transform and the original function. The only difference is the change in the sign of the exponent (change from i to $-i$) between the formulas for each of the two functions in terms of the other. We will also systematically use Latin letters in the time domain (or domain of real measurements) and Greek in the frequency domain.

Second, we must again examine *aliasing*. The discussion for periodic functions in Sec. 31.6 was in no way dependent upon the discrete frequencies of the function being sampled, but referred to the confusion due to the sampling of *any* frequency. Thus the results apply to the general function, not just to periodic functions, and the details need not be repeated here.

This aliasing is the reason why band-limited functions (functions whose frequencies all fall in a limited band) play such a leading role in the theory. Provided the frequencies are limited to the band

$$-\Omega < \sigma < \Omega$$

of width 2Ω , and provided we sample at an interval Δt between samples, then we must have (see Sec. 31.6)

$$2\Omega \Delta t < 1$$

to avoid aliasing. The maximum sampling interval $1/(2\Omega)$ is related to the folding, or Nyquist, frequency shown in Fig. 31.6.1. A common way of expressing this relation is to say that if we are to avoid aliasing, we must have at least two samples in the highest frequency present.

Third, the question naturally arises, Can we reconstruct a band-limited function from its samples if we obey the sampling restriction? While we will later (Sec. 34.6) give a somewhat more rigorous proof of the result, we now give an intuitive argument that it is possible.

In the Lagrange interpolation (Sec. 14.5) we introduced the $\pi_i(x)$ functions which were used to form (remember the i th factor is omitted)

$$\frac{\pi_i(x)}{\pi_i(x_i)} = \frac{(x - x_1)(x - x_2) \cdots (x - x_n)}{(x_i - x_1)(x_i - x_2) \cdots (x_i - x_n)}$$

This expression has the property that it takes on the value 1 at the i th sample point x_i and is 0 at all the other sample points. It was then used to form an interpolating polynomial

$$f(x) = \sum_{i=1}^N y_i \frac{\pi_i(x)}{\pi_i(x_i)}$$

through the given data.

The function

$$\frac{\sin \pi(t - k)}{\pi(t - k)}$$

which for equally spaced data has exactly the same properties, leads to the interpolating function

$$f(t) = \sum_{k=-\infty}^{\infty} f(k) \frac{\sin \pi(t - k)}{\pi(t - k)}$$

This is a formal expansion which will pass through all the sample points $(k, f(k))$ ($-\infty < k < \infty$), and we have ignored convergence questions.

We want to show that this function is band-limited and in particular has frequencies limited to the band $-1/2 < \sigma < 1/2$. It is easy to see by direct integration that

$$\int_{-1/2}^{1/2} e^{2\pi i \sigma t} d\sigma = \frac{e^{2\pi i t/2} - e^{-2\pi i t/2}}{2\pi i t} = \frac{\sin \pi t}{\pi t}$$

In this integral the function being integrated, $F(\sigma)$, is as shown in Fig. 34.3.1 and is band-limited ($-1/2 < \sigma < 1/2$).

Thus the function (see Fig. 34.5.2)

$$\frac{\sin \pi(k - t)}{\pi(k - t)}$$

is band-limited to the proper interval, using, as we have, $\Delta t = 1$. This result is the essence of the sampling theorem; given the proper equally spaced samples of a band-limited function $f(t)$, we can reconstruct the function from the samples by the above formula for $f(t)$.

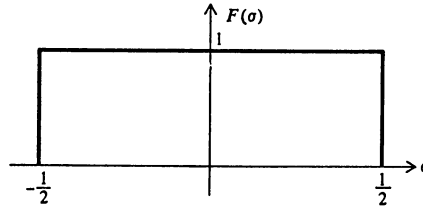


FIGURE 34.3.1

In analogy with the power spectrum of the Fourier-series power spectrum, the quantity

$$|F(\sigma)|^2 = F(\sigma)\overline{F(\sigma)}$$

is often called the *power spectrum* of the function $f(t)$. Instead of power spectrum the simple word *spectrum* is often used. One can regard a function $f(t)$ as if it were a light beam being broken up by a prism into a spectrum of colors—the Fourier integral breaks up the function into a spectrum of frequencies $F(\sigma)$. For a single frequency we often speak of a *spectral line*. Of course physically a pure spectral line does not occur, though very narrow lines are often produced by careful laboratory arrangements.

Fourth, as we shall show in Sec. 34.8, the effect of taking a finite-length sample of the function, in place of the true infinitely long function, is that the individual spectral lines are broadened in a specific way; and the longer the sample, the less the broadening.

The purpose of the rest of the chapter is to develop a little of the theory of the Fourier integral in order to give a more adequate presentation of the above results. We do not intend to become involved in the mathematical rigor since the class of functions that one is apt to use in computing is restricted to “well behaved” and excludes the pathological functions that are so dear to the hearts of the mathematically inclined.

For those whose skill in manipulating complex numbers is a bit rusty, the following problems are given for practice. If they cause any trouble, it would be well to study the topic before going on to the Fourier-integral theory since it makes use of them.

PROBLEMS

34.3.1 Prove $|e^{ix}| = 1$ (for real x).

34.3.2 Prove that the addition formulas of trigonometry follow from $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$.

34.3.3 If $1/(a+ib) = u+iv$, find u, v in terms of a and b .

34.3.4 If $u+iv = (a+ib)^{1/2}$, find u, v . (Watch the sign of the u term.)

34.3.5 If $c_k = \bar{c}_{-k}$, prove that $\sum_{k=-\infty}^{\infty} c_k e^{ikt}$ is real.

34.3.6 If m and k are integers, prove that

$$\int_{-\pi}^{\pi} e^{imx} e^{-ikx} dx = \begin{cases} 2\pi & m \neq k \\ 0 & m = k \end{cases}$$

34.3.7 If $f(t) = t$ ($-\pi < t < \pi$), find the Fourier coefficients in $f(t) = \sum_{k=-\infty}^{\infty} c_k e^{ikt}$.

34.3.8 If $f(t) = \int_{-\infty}^{\infty} F(\sigma) e^{2\pi i \sigma t} d\sigma$, find the transform of $f'(t)$, $f''(t)$, $f^{(k)}(t)$.

34.3.9 If

$$f(t) = \begin{cases} t & -\pi < t < \pi \\ 0 & \text{elsewhere} \end{cases}$$

find $F(\sigma)$.

34.3.10 If

$$F(\sigma) = \begin{cases} \sigma & -\pi < \sigma < \pi \\ 0 & \text{elsewhere} \end{cases}$$

find $f(t)$.

34.4 THE FOURIER INTEGRAL

The Fourier series for a periodic function in the interval $-N < t \leq N$ was given in Sec. 34.2, and if we eliminate the coefficients c_k , we will have, for the complex form

$$f(t) = \sum_{k=-\infty}^{\infty} \left[\int_{-N}^N f(t') e^{-(\pi i/N)kt'} dt' \right] e^{(\pi i/N)kt} \frac{1}{2N}$$

In order to approach the approximation of a nonperiodic function we will let the interval get longer and longer, that is, let $N \rightarrow \infty$. In the limit, the function will no longer be periodic since the interval of periodicity is the whole axis.

For the purpose of following what happens as $N \rightarrow \infty$ we set

$$\frac{1}{2N} = \Delta\sigma$$

and hence

$$\frac{k}{2N} = k \Delta\sigma = \sigma$$

We have therefore

$$f(t) = \sum_{k=-\infty}^{\infty} \left[\int_{-N}^N f(t') e^{-2\pi i \sigma t'} dt' \right] e^{2\pi i \sigma t} \Delta\sigma$$

Note that as N gets larger and larger the successive terms in the summation get closer and closer to each other; the exponentials are being packed closer and closer as $N \rightarrow \infty$. It is reasonable to suppose that in the limit the summation will go over into an integral, *provided* the function $f(t)$ is reasonably well behaved

$$f(t) = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(t') e^{-2\pi i \sigma t'} dt' \right] e^{2\pi i \sigma t} d\sigma$$

We now set

$$F(\sigma) = \int_{-\infty}^{\infty} f(t') e^{-2\pi i \sigma t'} dt'$$

and we have

$$f(t) = \int_{-\infty}^{\infty} F(\sigma) e^{2\pi i \sigma t} d\sigma$$

The function $F(\sigma)$ is said to be the Fourier transform of the function $f(t)$. The two functions $f(t)$ and $F(\sigma)$ have almost exactly reciprocal relationships to each other. The exception is in the sign of the exponent, $i \rightarrow -i$. Both functions contain the same information since each can be uniquely determined from the other. They present the same information in different forms; $f(t)$ is in the time domain, and $F(\sigma)$ is in the (complex) frequency domain. These two alternate, equivalent, views of the same information account for much of the value of the Fourier integral representation of the function.

34.5 SOME TRANSFORM PAIRS

The important relationship between the function $f(t)$ and its Fourier transform $F(\sigma)$ indicates that a table of the correspondences would be useful, and such tables have been made [6, 9]. We propose to develop only a very few elementary transforms that we will need for further work, and leave the more general results to books which concentrate on the Fourier integral.

As a first example consider the band-limited function (see Fig. 34.5.1)

$$F(\sigma) = \begin{cases} \frac{1}{2\Omega} & \text{for } |\sigma| < \Omega \\ 0 & \text{for } |\sigma| > \Omega \end{cases}$$

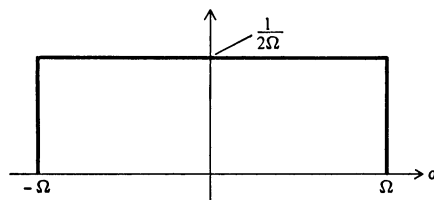


FIGURE 34.5.1

Then, as before,

$$\begin{aligned} f(t) &= \int_{-\infty}^{\infty} F(\sigma) e^{2\pi i \sigma t} d\sigma = \frac{1}{2\Omega} \int_{-\Omega}^{\Omega} e^{2\pi i \sigma t} d\sigma \\ &= \frac{1}{2\Omega} \frac{e^{2\pi i \sigma t}}{2\pi i t} \Big|_{-\Omega}^{\Omega} = \frac{e^{2\pi i \Omega t} - e^{-2\pi i \Omega t}}{2i} \frac{1}{2\Omega \pi t} \\ &= \frac{\sin 2\pi \Omega t}{2\pi \Omega t} \end{aligned}$$

From this it follows immediately that

$$F(\sigma) = \int_{-\infty}^{\infty} \frac{\sin 2\pi\Omega t}{2\pi\Omega t} e^{-2\pi i\sigma t} dt$$

The function

$$\frac{\sin 2\pi\Omega t}{2\pi\Omega t} = 1 - \frac{(2\pi\Omega t)^2}{3!} + \frac{(2\pi\Omega t)^4}{5!} - \dots$$

occurs frequently and Fig. 34.5.2 shows how it looks.

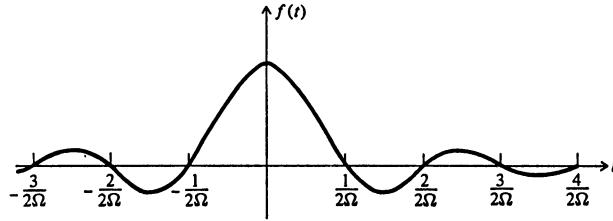


FIGURE 34.5.2
The band-limited function $\sin(2\pi\Omega t/2\pi\Omega t)$.

We see that there is a main lobe extending from $-1/2(\Omega)$ to $1/2(2\Omega)$ and that beyond this the function oscillates with the zeros equally spaced while the amplitude of the oscillation slowly decays toward zero. The larger the Ω , the narrower the spike at the origin.

As a second illustration of a relationship between Fourier transforms, suppose that we already know a transform pair $f(t)$ and $F(\sigma)$; that is, we know

$$f(t) = \int_{-\infty}^{\infty} F(\sigma) e^{2\pi i\sigma t} d\sigma$$

We ask, “What function $f_1(t)$ corresponds to $F(\sigma)e^{2\pi i\sigma y}$?” We have from the definition

$$\begin{aligned} f_1(t) &= \int_{-\infty}^{\infty} F(\sigma) e^{2\pi i\sigma y} \cdot e^{2\pi i\sigma t} d\sigma \\ &= \int_{-\infty}^{\infty} F(\sigma) e^{2\pi i\sigma(y+t)} d\sigma \end{aligned}$$

Clearly $y + t$ here plays the role of t above, so that

$$f_1(t) \equiv f(y + t)$$

Thus the effect of the exponential multiplier $e^{2\pi i\sigma y}$ is to *shift* the argument of the transform y units. This result is often known as the “shifting theorem.”

PROBLEMS

34.5.1 Find the transform of

$$f(t) = \begin{cases} 1 - |t| & |t| \leq 1 \\ 0 & |t| \geq 1 \end{cases}$$

34.5.2 Find the transform of

$$f(t) = \begin{cases} t & \text{for } |t| \leq 1 \\ 0 & \text{for } |t| > 1 \end{cases}$$

34.5.3 What are the transforms of Probs. 34.5.1 and 34.5.2 if the origin is shifted k units?34.5.4 Note that the product σy does not change if $\sigma \rightarrow k\sigma$ and $y \rightarrow y/k$. Deduce the "stretch theorem" for Fourier integrals.

34.6 BAND-LIMITED FUNCTIONS AND THE SAMPLING THEOREM

In Sec. 34.3 we gave a formal intuitive justification of the sampling theorem. Because of the importance of this theorem in computing we shall give another, somewhat more mathematical, derivation though it is still not mathematically rigorous with all the pathological details examined carefully.

The central idea of the sampling theorem is that a band-limited function $f(t)$, extending from $t = -\infty$ to $t = +\infty$, is sampled at equally spaced points with a spacing such that at least two samples occur in the highest frequency present. The restriction to band-limited functions is very reasonable since many physical situations are approximately band-limited. However, it is necessary to include a few words of caution. If $f(t)$ is band-limited, then the mathematical model says that it cannot be "time-limited" in the sense that outside some interval, no matter how large, $f(t)$ cannot vanish identically. In particular, if the band-limited function represents an electrical current, then the current must have been flowing for all past time and will flow for all future time. Correspondingly, if $f(t)$ is time-limited, then it cannot be band-limited, and there must be arbitrarily high frequencies present. Evidently the mathematical model should not be pushed too hard when applying it to the real world; it is only a very useful approximation to reality.

In the mathematical treatment, the band-limited aspect arises from the effect of sampling we have called aliasing. If aliasing is to be avoided, then we must limit ourselves to frequencies inside the Nyquist, or folding, frequency (for equally spaced data).

To derive the sampling theorem, suppose we are given the function $F(\sigma)$ in the band $-\Omega < \sigma < \Omega$ and outside the interval $F(\sigma)$ is zero. We first replace it by

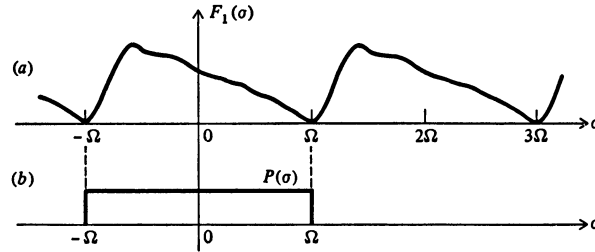


FIGURE 34.6.1.
(a) $F_1(\sigma)$ is periodic. (b) $P(\sigma)$ is the rectangular pulse.

the function $F_1(\sigma)$ which coincides with $F(\sigma)$ inside the interval and is periodic outside as shown in Fig. 34.6.1.

We have for $F(\sigma)$

$$F(\sigma) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i \sigma t} dt$$

$$f(t) = \int_{-\infty}^{\infty} F(\sigma) e^{2\pi i \sigma t} d\sigma = \int_{-\Omega}^{\Omega} F(\sigma) e^{2\pi i \sigma t} d\sigma$$

For $F_1(\sigma)$ we have, since it is periodic, the Fourier-series expansion

$$F_1(\sigma) = \sum_{k=-\infty}^{\infty} c_k e^{(\pi i/\Omega)k\sigma}$$

where

$$c_k = \frac{1}{2\Omega} \int_{-\Omega}^{\Omega} F_1(\sigma) e^{-(\pi i/\Omega)k\sigma} d\sigma$$

From the equation for $f(t)$ this is exactly

$$c_k = \frac{1}{2\Omega} f\left(\frac{-k}{2\Omega}\right)$$

Now consider $P(\sigma)$ a rectangular pulse (see Fig. 34.6.1b)

$$P(\sigma) = \begin{cases} \frac{1}{2\Omega} & \text{for } |\sigma| < \Omega \\ 0 & \text{for } |\sigma| > \Omega \end{cases}$$

By our previous result (Sec. 34.5) the transform of $P(\sigma)$ is the band-limited function

$$p(t) = \frac{\sin 2\pi\Omega t}{2\pi\Omega t}$$

From the definitions of $F_1(\sigma)$ and $P(\sigma)$ it is clear that the original band-limited function $F(\sigma)$ is

$$F(\sigma) = F_1(\sigma)P(\sigma)2\Omega$$

Substituting for $F_1(\sigma)$, we get

$$F(\sigma) = \sum_{k=-\infty}^{\infty} c_k e^{(\pi i/\Omega)k} P(\sigma)2\Omega$$

Using the values of the c_k , we find

$$F(\sigma) = \sum_{k=-\infty}^{\infty} f\left(\frac{-k}{2\Omega}\right) P(\sigma) e^{(\pi i/\Omega)k\sigma}$$

We now transform back to the time domain and apply the shifting theorem

$$\begin{aligned} f(t) &= \int_{-\infty}^{\infty} F(\sigma) e^{2\pi i \sigma t} d\sigma \\ &= \sum_{k=-\infty}^{\infty} f\left(\frac{-k}{2\Omega}\right) \int_{-\infty}^{\infty} P(\sigma) e^{\pi i/\Omega k} e^{2\pi i \sigma t} d\sigma \\ &= \sum_{k=-\infty}^{\infty} f\left(\frac{-k}{2\Omega}\right) \frac{\sin 2\pi\Omega(t + k/2\Omega)}{2\pi\Omega(t + k/2\Omega)} \end{aligned}$$

Now replace k by $-k$

$$f(t) = \sum_{k=-\infty}^{\infty} f\left(\frac{k}{2\Omega}\right) \frac{\sin \pi(2\Omega t - k)}{\pi(2\Omega t - k)}$$

and we have the sampling theorem.

At the folding frequency itself we cannot reconstruct the function from the samples, since if $\Delta t = 1$ and $f(t) = \sin \pi t$, then all the samples would be zero and the sampling theorem would give $f(t) \equiv 0$.

PROBLEMS

34.6.1 Write the function that is band-limited $-1/2 < \sigma < 1/2$ and has $f(0) = f(1) = f(2) = 1$ and all other $f(k) = 0$.

34.7 THE CONVOLUTION THEOREM

A fundamental relation involving Fourier transforms is known as *the convolution theorem*. Suppose that we have two functions $f(t)$ and $g(t)$. The convolution $h(t)$ of $f(t)$ with $g(t)$ is defined as

$$h(t) = \int_{-\infty}^{\infty} f(s)g(t-s) ds$$

Note that for a fixed t we can write $t - s = s'$, and the above equation becomes

$$h(t) = \int_{-\infty}^{\infty} f(t - s')g(s') ds'$$

Thus the convolution of $f(t)$ with $g(t)$ is the same as the convolution of $g(t)$ with $f(t)$.

We now ask, "What is the Fourier transform of the convolution which we have labeled $h(t)$?" By definition, the transform of $h(t)$ is

$$H(\sigma) = \int_{-\infty}^{\infty} h(t)e^{-2\pi i\sigma t} dt$$

Substituting for $h(t)$, we get

$$\begin{aligned} H(\sigma) &= \int_{-\infty}^{\infty} f(s)e^{-2\pi i s\sigma} \left[\int_{-\infty}^{\infty} g(t-s)e^{-2\pi i(t-s)\sigma} dt \right] ds \\ &= \int_{-\infty}^{\infty} f(s)e^{-2\pi i s\sigma} G(\sigma) ds \\ &= F(\sigma)G(\sigma) \end{aligned}$$

Thus we have the result: *the transform of the convolution of two functions is the product of their transforms.*

The above formal derivation proves the result for two time functions, and Prob. 34.7.1 covers the case of the convolution of two frequency functions.

Frequently the convolution of a function with itself is of interest. We have

$$h(t) = \int_{-\infty}^{\infty} f(s)f(t-s) ds$$

Applying the Fourier transform twice, we get

$$h(t) = \int_{-\infty}^{\infty} F^2(\sigma)e^{-2\pi i\sigma t} d\sigma$$

and for $t = 0$

$$h(0) = \int_{-\infty}^{\infty} f(s)f(-s) ds = \int_{-\infty}^{\infty} F^2(\sigma) d\sigma$$

PROBLEMS

34.7.1 Prove the formula for the convolution of two frequency functions.

34.7.2 Derive the last formula starting with the frequency functions.

34.7.3 Calculate the convolution of a rectangular pulse of unit area with itself.

34.8 THE EFFECT OF A FINITE SAMPLE SIZE

Many time functions $f(t)$ can be regarded as infinitely long signals (in time). Of necessity we must take a finite length sample. Thus in astronomy we can observe a pulsar or a Cepheid variable for a finite length of time only. What does this limitation do to the original signal? We can regard the limitation as being equivalent to multiplication by the rectangular pulse

$$p(t) = \begin{cases} \frac{1}{2T} & |t| < T \\ 0 & |t| > T \end{cases}$$

Thus what we see is not $f(t)$ but rather $f_1(t)$, defined as

$$f_1(t) = p(t)f(t)2T$$

The Fourier transform is, by the convolution theorem,

$$F_1(\sigma) = 2T \int_{-\infty}^{\infty} F(\sigma_1) \frac{\sin 2\pi T(\sigma_1 - \sigma)}{2\pi T(\sigma_1 - \sigma)} d\sigma_1$$

Thus the transform of what we see is the convolution of the true transform $F(\sigma)$ with the function

$$2T \frac{\sin 2\pi T(\sigma_1 - \sigma)}{2\pi T(\sigma_1 - \sigma)}$$

To understand this, consider the situation in which the signal is a single frequency, a spike in the spectrum. This spike is convolved with the above function, and therefore we will see the above function. This is the usual $(\sin x)/x$ function with the amplitude growing like $2T$ and the half-width to the nearest zero contracting like $1/2T$. If instead of a single spike in the spectrum we have many lines, each will be smeared out by the same $(\sin x)/x$, and in the limit of a continuous spectrum we will get the above convolution integral. In an optical analogy this smearing out corresponds to the idea of resolving power—the longer the time signal is observed, the better we can resolve adjacent spectral lines. While it is intuitively evident that something like this should happen, we have exhibited the actual dependence on T . Thus we can now understand what happens to a function when we chop out a piece of it to observe closely, and this limits what we can hope to find out from the sample.

35

A SECOND LOOK AT POLYNOMIAL APPROXIMATION—FILTERS

35.1 PURPOSE OF CHAPTER

The purpose of this chapter is to look at a number of things we did earlier in Part II, using the tools we have just developed, so that we may see the results in a different light. We shall look at only the formulas which used equally spaced points, since this is an essential part of the Fourier-approximation approach. It is more convenient to use the notation

$$y(t) = e^{2\pi i \sigma t} \equiv e^{i\omega t} \quad \omega = 2\pi\sigma$$

than it is to use the sines and cosines.

It is important to notice that since the formulas are linear, the result of substituting $e^{i\omega t}$ into a formula is that the complex exponential depending on t will factor out and leave a fixed multiplier independent of t . This is often called *filtering*, and the formula is called a *filter*. The word *filter* arose in engineering practice and, like *power spectrum*, persists in the language even when the method is applied outside the original field.

35.2 ROUND OFF NOISE

Roundoff is an essential, inescapable effect in scientific computing, and a roundoff theory was developed in Chap. 10. The use of the difference table there was justified on the basis of local polynomial approximation. Let us now ask what the difference process does to a single complex sinusoid (where for convenience we have scaled σ , or ω , to make $\Delta t = h = 1$)

$$y(t) = e^{2\pi i \sigma t} = e^{i\omega t}$$

We have the linear formula

$$\Delta y_n = y_{n+1} - y_n = e^{i\omega(t+1)} - e^{i\omega t}$$

hence

$$\Delta y = e^{i\omega t} \cdot e^{i\omega/2} [e^{i\omega/2} - e^{-i\omega/2}] = \left[2i \sin \frac{\omega}{2} \right] (e^{i\omega/2}) e^{i\omega t}$$

Thus differencing multiplies the function by a factor of size

$$2 \sin \frac{\omega}{2} = 2 \sin \pi \sigma$$

See Fig. 35.2.1. The second difference naturally has a factor of size

$$\left(2 \sin \frac{\omega}{2} \right)^2$$

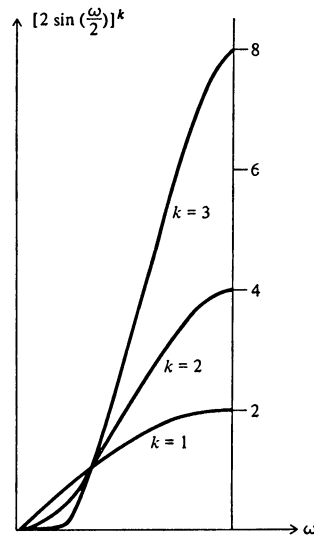


FIGURE 35.2.1

and the k th difference has a factor of size

$$\left(2 \sin \frac{\omega}{2}\right)^k = (2 \sin \pi \sigma)^k$$

As a check at $\sigma = 1/2$, which is the Nyquist sampling rate,

$$y(t) = e^{\pi i t} = (-1)^t$$

and the k th difference is, as it should be, of size 2^k .

t	$(-1)^t$	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6	Δ^7
0	+1							
1	-1	-2						
2	+1	+2	+4					
3	-1	-2	-4	-8				
4	+1	+2	+4	+8	+16			
5	-1	-2	-4	-8	-16	-32		
6	+1	+2	+4	+8	+16	+32	+64	
7	-1	-2	-4	-8	-16	-32	-64	-128

An examination of the figure shows that at $1/3$ the Nyquist sampling rate the multiplying factor is exactly 1. Thus below $1/3$ the Nyquist rate, the process of repeated differencing (the usual difference table) tends to dampen (decrease the amplitude of) the frequency, while above this rate it amplifies the amplitude of the frequency. We are led, in this fashion, to *identify noise with high frequency*, and signal with low frequency, with $1/3$ the Nyquist rate being the boundary between them.

And this is not unreasonable. Roundoff at a point resembles suddenly adding a small square pulse to the function, and we have already seen that the coefficients of a square pulse drop off like $1/n$. The difference table will rapidly amplify the higher frequencies and dampen the lower ones of each roundoff that occurs.

Thus we see that exactly the same computation, the difference table in this case, can be seen in a different light when we ask what the process of differencing does to the various frequencies that may be in the function being differenced. Although the computing is the same, the roundoff model is different and often more suggestive of physical reality.

35.3 DERIVATIVES

Let us look at the simple formula for finding a derivative from the differences of a function. This resembles differencing. We use the central difference formula

$$f'_k = \frac{f_{k+1} - f_{k-1}}{2h} \quad h = \text{spacing}$$

and again ask what it does to a single frequency

$$e^{2\pi i \sigma t} = e^{i\omega t}$$

At the k th sample point $t_k = hk = k$ (since we are assuming $h = 1$)

$$f_k = e^{i\omega k}$$

We compute

$$f'_k = \frac{e^{i\omega(k+1)} - e^{i\omega(k-1)}}{2} = e^{i\omega k}(i \sin \omega)$$

On the other hand, the correct value is

$$f'_k = i\omega e^{i\omega k}$$

The ratio of the computed answer to the true answer is

$$\frac{i \sin \omega e^{i\omega k}}{i\omega e^{i\omega k}} = \frac{\sin \omega}{\omega}$$

From the plot of $(\sin \omega)/\omega$ in Sec. 34.5 we see that the difference formula clearly underestimates all frequencies except $\omega = 0$ (which is often called dc for direct current). As one would expect, when $\omega = \pi$, that is, when $\sigma = 1/2$, we get an estimate of zero for the derivative ($\sigma = 1/2$ is the folding, or Nyquist, frequency).

Similarly for the second derivative,

$$f''_k = \frac{f_{k+1} - 2f_k + f_{k-1}}{h^2} \quad h = 1$$

we get the estimate

$$f''_k = e^{i\omega k}(e^{i\omega} - 2 + e^{-i\omega}) = -e^{i\omega k}[2(1 - \cos \omega)]$$

The correct answer is

$$f''_k = -\omega^2 e^{i\omega k}$$

so that the ratio of the computed to true is

$$\frac{2(1 - \cos \omega)}{\omega^2} = \frac{\sin^2 \omega/2}{(\omega/2)^2} = \left[1 - \frac{\omega^2}{24} + \cdots\right]^2$$

which again underestimates the quantity except at dc. At the folding frequency the ratio is $4/\pi^2$.

Since, as we already knew, the estimation of derivatives from computed or tabulated values is dangerous, we usually try to avoid them when possible.

Many times the problem of estimating a derivative can be avoided by proper analysis. As an example, suppose we have data f_k and we know that theoretically $f(t)$ satisfies a second-order differential equation of the form

$$f'' = H(f, t)$$

Using this, we can go from values of $f(t)$ to values of $f''(t)$ and can integrate the latter to obtain $f'(t)$. We frequently prefer integration to differentiation, provided that we are not operating on so long a run of data that a slow, steady drift in the integration process would ultimately produce a large error.

Numerous other tricks have been found in special cases, but there appears to be no general theory of when and how derivatives can be avoided.

For differentiation we have, also, Lanczos' method of differentiation, Sec. 32.9, but this is based on Fourier approximation, not on polynomial approximation.

35.4 INTEGRATION—A FIRST LOOK

We will take up a detailed examination of integration along with the treatment of differential equations in the next chapter, but it is worthwhile to first look at what the frequency approach tells us about simple formulas such as the composite trapezoid and midpoint formulas. We have for the trapezoid rule the formula

$$\frac{1}{2}f_0 + f_1 + f_2 + \cdots + f_{n-1} + \frac{1}{2}f_n$$

($h = 1$) and again suppose that

$$f_k = e^{2\pi i \sigma k} = e^{i\omega k}$$

We get immediately

$$\begin{aligned} \frac{1}{2} + e^{i\omega} + e^{2i\omega} + \cdots + e^{i(n-1)\omega} + \frac{1}{2}e^{i\omega n} &= \frac{1 - e^{i(n+1)\omega}}{1 - e^{i\omega}} - \frac{1}{2}(1 + e^{i\omega n}) \\ &= \frac{(1 + e^{i\omega})(1 - e^{i\omega n})}{2(1 - e^{i\omega})} \end{aligned}$$

The true answer is, of course,

$$\int_0^n e^{i\omega t} dt = \frac{e^{i\omega n} - 1}{i\omega}$$

The ratio of the computed to true is therefore

$$\begin{aligned} \frac{-(1 + e^{i\omega}) \cdot i\omega}{2(1 - e^{i\omega})} &= \frac{e^{i\omega/2} + e^{-i\omega/2}}{e^{i\omega/2} - e^{-i\omega/2}} \frac{i\omega}{2} \\ &= \frac{\cos \omega/2}{\sin \omega/2} \frac{\omega}{2} = \frac{\omega}{2} \cot \frac{\omega}{2} \\ &= 1 - \frac{\omega^2}{12} - \frac{\omega^4}{720} - \dots \end{aligned}$$

We see that the error behaves like ω^2 , and for low frequencies the formula is quite accurate.

For the midpoint formula we get, correspondingly,

$$\begin{aligned} f_{1/2} + f_{3/2} + \dots + f_{(2n-1)/2} &= e^{i\omega/2} + e^{3i\omega/2} + \dots + e^{(2n-1)i\omega/2} \\ &= e^{i\omega/2} \frac{1 - e^{i\omega n}}{1 - e^{i\omega}} \end{aligned}$$

The ratio of computed to true is

$$\begin{aligned} \frac{e^{i\omega n} - 1}{e^{i\omega/2} - e^{-i\omega/2}} \frac{i\omega}{e^{i\omega n} - 1} &= \frac{\omega/2}{\sin(\omega/2)} \\ &\approx 1 + \frac{\omega^2}{24} + \frac{7\omega^4}{5760} + \dots \end{aligned}$$

which is by now a familiar function. This differs from the trapezoid rule by the omission of the factor $\cos(\omega/2)$ and already does better at the lower frequencies by a factor of 2.

35.5 SMOOTHING, AN EXAMPLE OF DESIGN

In the polynomial treatment of formulas we tried to avoid the topics of both differentiation and smoothing, though a little of each crept in. Least-squares smoothing, by fitting a local polynomial and using the value from the polynomial as the smoothed value, was an example we gave of smoothing. In the simplest case we fit by a constant value, and found in Sec. 27.7 (Prob. 27.7.2) that this led to the average value.

For three points we get

$$\frac{f_{k-1} + f_k + f_{k+1}}{3} = g_k = \text{smoothed value}$$

What happens to our complex sinusoid? Let

$$\begin{aligned} f_k &= e^{2\pi i \omega k} \\ g_k &= e^{i\omega k} \frac{e^{-i\omega} + 1 + e^{i\omega}}{3} \\ &= e^{i\omega k} \left[\frac{1 + 2 \cos \omega}{3} \right] \end{aligned}$$

The ratio of smoothed to true is

$$\frac{1 + 2 \cos \omega}{3} = \frac{\sin(3\omega/2)}{3 \sin(\omega/2)}$$

In the spectrum, this factor is

$$\frac{\sin^2(3\omega/2)}{9 \sin^2(\omega/2)}$$

We have already learned that the sampling frequency is directly connected with the folding frequency in the spectrum. Suppose we have a signal that is band-limited and has a maximum frequency of 10 Hz. This forces us to 20 samples per second to avoid aliasing.

But suppose that we are interested only in frequencies less than 5 Hz. We dare not sample at 10 Hz, since then the part of the spectrum above 5 Hz would be aliased down into the interval of interest.

The filter in this section suggests a way of eliminating the upper half of the spectrum. As the filter stands, it reduces the frequencies by at least a factor of 1/9.

We can do better by following the first filter with a second smoothing filter, smoothing by 4s,

$$\frac{g_k + g_{k+1} + g_{k+2} + g_{k+3}}{4} \equiv h_k$$

Then h_k has this effect on $e^{2\pi i \omega t} = e^{i\omega t}$

$$\begin{aligned} h_k &= \frac{e^{i\omega k}}{4} (1 + e^{i\omega} + e^{2i\omega} + e^{3i\omega}) \\ &= \frac{e^{i\omega k}}{4} \frac{1 - e^{4i\omega}}{1 - e^{i\omega}} = \frac{e^{i\omega(k+2)}}{4e^{i\omega/2}} \frac{e^{2i\omega} - e^{-2i\omega}}{e^{i\omega/2} - e^{-i\omega/2}} \\ &= \frac{e^{i(k+3/2)\omega}}{4} \frac{\sin 2\omega}{\sin \omega/2} \end{aligned}$$

Hence the spectrum is affected in magnitude by

$$\left(\frac{\sin 2\omega}{4 \sin \omega/2} \right)^2 = \left(\frac{\sin 4\pi\sigma}{4 \sin \pi\sigma} \right)^2$$

which has zeros at $\sigma = 1/4, 1/2, \dots$

The result of the two filters, the smoothing by 3s followed by smoothing by 4s, is the filter

$$h_k = \frac{f_k + 2f_{k+1} + 3f_{k+2} + 3f_{k+3} + 2f_{k+4} + f_{k+5}}{12}$$

and its ability to reject frequencies in the upper half of the spectrum is very good. See Fig. 35.5.1 where we have plotted the log of the spectrum versus σ .

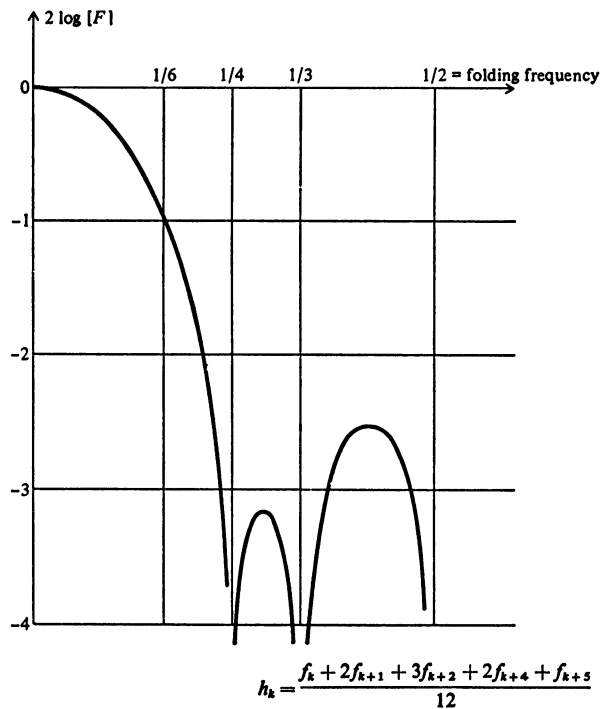


FIGURE 35.5.1
Response of the filter.

$$h_k = \frac{f_k + 2f_{k+1} + 3f_{k+2} + 2f_{k+4} + f_{k+5}}{12}$$

Having smoothed, we can safely drop alternate terms h_k , keeping, say, h_{2k} , and analyze the result without any serious fear of the resulting aliasing (the folding frequency is now at 5 Hz) *unless* there was a very large amount of “power” in the upper half of the spectrum compared to that in the lower half. When we finally use the results, we must make due allowance for the effect of filtering on the frequencies up to 5 Hz. Thus the filtering action of the filters allows us to subsample the function without having aliasing troubles. The accuracy for frequencies between 1/3 and 1/2 of the original folding frequency is not very good.

It is easy to compute the effect of smoothing by averaging m consecutive values, namely

$$\frac{\sin(\pi m \sigma)}{m \sin(\pi \sigma)}$$

35.6 LEAST-SQUARES SMOOTHING

In Sec. 27.7 we examined the smoothing of data by passing a least-squares quadratic through five points, and then using the value of the quadratic as the smoothed value at the midpoint. This resulted in the smoothing formula

$$g = \frac{-3y_{-2} + 12y_{-1} + 17y_0 + 12y_1 - 3y_2}{35}$$

or

$$g = y_0 - \frac{3}{35} \Delta^2 y_{-2}$$

Using the usual substitution of

$$y = e^{i\omega t}$$

we find in the frequency domain

$$G(\omega) = \frac{17 + 24 \cos \omega - 6 \cos 2\omega}{35}$$

In the spectrum we would, of course, use the squares of the tabled values of $G(\omega)$ (Table 35.6).

If we smoothed more drastically by fitting a constant, then we would have the least-squares smoothing by 5s. These two curves are shown in Fig. 35.6.1, along with a curve from the next section on Chebyshev smoothing.

Table 35.6

ω	$G(\omega)$
0	1.000
$\pi/6$	0.995
$2\pi/6$	0.915
$3\pi/6$	0.658
$4\pi/6$	0.228
$5\pi/6$	-0.194
$6\pi/6$	-0.371

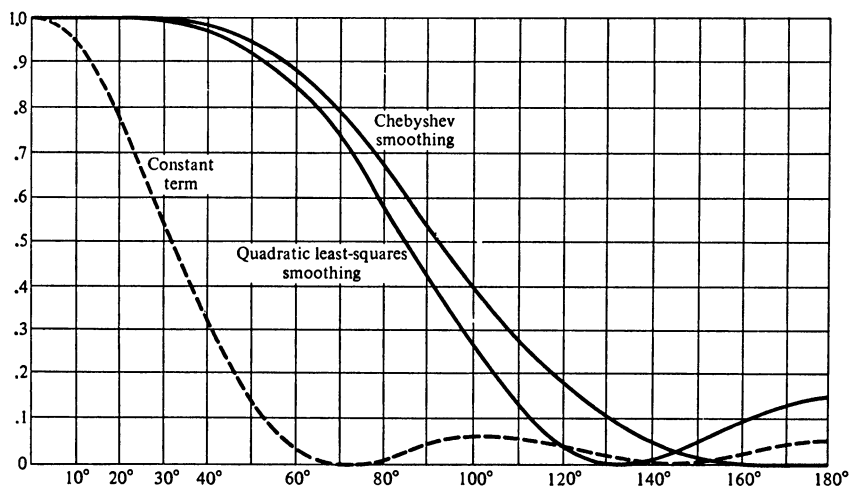


FIGURE 35.6.1
Least-squares filters using five points.

PROBLEMS

35.6.1 Plot the curve for smoothing by 3s on Fig. 35.6.1. Discuss the results.

35.7 CHEBYSHEV SMOOTHING

After examining least-squares smoothing on 5 points $(-2, -1, 0, 1, 2)$ from the frequency point of view, it is natural to ask the same question about Chebyshev smoothing. What does it do to the signal? We proceed as follows:

- 1 Find the quartic through the 5 points. To do this, assume that the quartic is

$$y(x) = a + bx + cx^2 + dx^3 + ex^4$$

and impose the five conditions. We easily see that

$$a = y_0$$

$$c = \frac{1}{24}(-y_{-2} + 16y_{-1} - 30y_0 + 16y_1 - y_2)$$

$$e = \frac{1}{24}(y_{-2} - 4y_{-1} + 6y_0 - 4y_1 + y_2)$$

- 2 We next observe that the particular quartic

$$T(x) = 2x^4 - 8x^2 + 3$$

is equal-ripple *at the sample points* since

$$\begin{aligned}T(-2) &= 3 \\T(-1) &= -3 \\T(0) &= 3 \\T(1) &= -3 \\T(2) &= 3\end{aligned}$$

- 3 From the quartic fitting of the data we subtract a suitable multiple of $T(x)$, so that we have a cubic

$$g(x) = y(x) - \frac{e}{2} T(x) = \text{cubic in } x$$

and therefore we have a Chebyshev fit to the data. The smoothed result is

$$\begin{aligned}g(0) &= a - \frac{e}{2} T(0) = \text{smoothed value} \\&= \frac{1}{16} (-y_{-2} + 4y_{-1} + 10y_0 + 4y_1 - y_2) \\&= y_0 - \frac{1}{16} \Delta^4 y_{-2}\end{aligned}$$

- 4 In the smoothing formula we put $y_n = e^{i\omega n}$ to get the effect on the frequencies, the “transfer function”

$$\frac{1}{8} (5 + 4 \cos \omega - \cos 2\omega)$$

- 5 To find the zeros of this function we set $\cos \omega = z$

$$\begin{aligned}5 + 4z - (2z^2 - 1) &= 0 \\z^2 - 2z - 3 &= 0 \\(z + 1)(z - 3) &= 0 \\z &= -1, 3\end{aligned}$$

Only $z = \cos \omega = -1$ leads to real frequencies, and this is $\omega = \pi$.

- 6 We tabulate the transfer function (Table 35.7)

Table 35.7

ω	Transfer function
0	1.0000
$\pi/6$	0.9955
$2\pi/6$	0.9375
$3\pi/6$	0.7500
$4\pi/6$	0.4375
$5\pi/6$	0.1295
$6\pi/6$	0.0000

and plot it on the chart with the least-squares smoothing functions for comparison purposes.

The comparisons of the smoothing formulas as shown in Fig. 35.6.1 speak for themselves.

PROBLEMS

35.7.1 Discuss smoothing by the family of formulas $g_n = y_n + \alpha \Delta^4 y_{n-2}$

35.8 THE FOURIER INTEGRAL

We have derived a number of expressions for the ratio of the computed to the true value for a sinusoid of a given frequency. Since the frequency was arbitrary, we have, in effect, the result for all frequencies. Thus we are in a position to apply the Fourier integral.

Suppose the function to which we wish to apply the formula has the Fourier transform

$$f(t) = \int_{-\infty}^{\infty} F(\sigma) e^{2\pi i \sigma t} d\sigma$$

Thus $F(\sigma)$ gives the amplitude of the frequency σ . Let the ratio of the calculated to true values be labeled

$$G(\sigma)$$

Then the result of amplitude $F(\sigma)$ is

$$F(\sigma)G(\sigma)$$

and we have that the output, or computed value for $f(t)$, is

$$\hat{f}(t) = \int_{-\infty}^{\infty} F(\sigma)G(\sigma) e^{2\pi i \sigma t} d\sigma$$

The process then is: given the function $f(t)$, we find the transform $F(\sigma)$ and then transform back the product $F(\sigma)G(\sigma)$ to get the effect of the formula, where $G(\sigma)$ is the ratio we have been computing.

35.9 SUMMARY

In summary, we have looked at a number of formulas in Part II that we first found by polynomial approximation. In each case the frequency approach to the same formula has given new understanding of what the formula does. It is important to realize that the errors in polynomial approximation are usually expressed in terms of high-order derivatives which are very rarely available before the computation, and often not even during the computation (consider, for example, Gaussian quadrature). Fourier approximation, on the other hand, uses frequencies which can often be estimated from the physical source of the problem. They can also be estimated from the rate of convergence of the Fourier representation of the function, which is controlled by the singularities of the function along the real axis where the measurements are made. The size of the derivatives in the polynomial case are controlled by the singularities in the complex part of the plane, where there are no measurements and little intuition to be used. Furthermore, the step size used in the computation can be related to the frequencies present in the problem itself. Thus the Fourier approach lives up to the motto of the book, *The Purpose of Computing Is Insight, Not Numbers*.

36.1 INTRODUCTION

Now that we have developed suitable tools, we can look more closely at the important computing problems of integration and the numerical solution of ordinary differential equations. The main new tool we have is the Fourier integral. It is important to notice that in the polynomial case we rated a formula by being exactly true, within roundoff, for $1, x, x^2, \dots, x^{m-1}$ and measured the error by the next power E_m . Similarly, in the Fourier series we made the formula exact for a finite sequence of functions $1, \cos x, \sin x, \dots, \cos(n-1)x, \sin(n-1)x, \cos nx$. But with the Fourier integral we look at the error as a function of the noncountable number of frequencies in the Nyquist interval, and we have been plotting the ratio of the computed result to the true answer as a function of the (continuous) frequency σ (or ω).

For a while we will use this new method of evaluating old formulas, and variants of the method, but any new method of evaluation gradually brings new methods of design. These we will look at only briefly.

There is a second important idea that we have to discuss. It goes under the fancy names of *recursive* and *nonrecursive* filter design. The difference is simple.

In a nonrecursive design only the function (input) values are used, while for a recursive design both input and output can be used (which means a feedback loop and possible instability). Clearly, we have used a recursive design in the past when we studied integration because there we used both the integrand and the solution values to predict the next value. And in any case the distinction is not all that clear. In Sec. 35.4 we analyzed the trapezoid rule as a nonrecursive formula, and in the next section we will analyze it as a recursive formula. Because recursive formulas use more information, generally they give better results—but because of feedback they also require more care in their design.

36.2 SIMPLE RECURSIVE INTEGRATION FORMULAS

The trapezoid rule may be looked at as an open-loop nonrecursive formula, as we did in Sec. 35.4, or else as a recursive formula in the form

$$y_{n+1} = y_n + \frac{h}{2} [f_n + f_{n+1}] \quad \text{Set } h = 1$$

We set

$$\begin{aligned} f_k &= A_I e^{i\omega k} \\ y_k &= A_O e^{i\omega k} \end{aligned}$$

where A_I is the input amplitude and A_O is the output amplitude. We get

$$A_O(e^{i\omega} - 1) = \frac{A_I}{2} (e^{i\omega} + 1)$$

Hence

$$\frac{A_O}{A_I} = \frac{1}{2} \frac{e^{i\omega} + 1}{e^{i\omega} - 1} = \frac{1}{2i} \frac{\cos(\omega/2)}{\sin(\omega/2)}$$

The true ratio is $1/(i\omega)$; hence the ratio of the computed to true answers is, as before,

$$\frac{\omega}{2} \cot \frac{\omega}{2}$$

Similarly, for the midpoint formula the recursive form is

$$y_{n+1} = y_n + f_{n+1/2}$$

We get

$$\begin{aligned} A_O(e^{i\omega} - 1) &= A_I e^{i\omega/2} \\ \frac{A_O}{A_I} &= \frac{1}{e^{i\omega/2} - e^{-i\omega/2}} = \frac{1}{2i \sin(\omega/2)} \end{aligned}$$

and the usual ratio of computed (ratio) to true (ratio) is

$$\frac{\omega/2}{\sin(\omega/2)}$$

For these two cases the recursive form is easier to handle in the analyses of what they do.

Let us pause to comment of this method in more detail so that we will be clear about what we are doing and what we are getting from the method. Suppose we were operating on a real sinusoid $A \sin 2\pi\sigma t$. If we were to measure the quality of the formula by the size of the error itself, then we would need to know whether $A = 1$ or $A = 1,000$ before stating how serious an error really is. It is much more reasonable to measure, as we have done repeatedly in the book, the *relative error* rather than the absolute error. But as we have noted in the past, near a zero the relative error is very misleading. However, when we use the complex sinusoid $Ae^{2\pi i\sigma t}$, as we do, for real σ and t , the modulus of the complex sinusoid is fixed in size

$$|Ae^{2\pi i\sigma t}| = |A|$$

and there are no zeros; hence the relative error now provides a very satisfactory measure of accuracy.

We have been cavalier about the phase angle. We can always suppose that it is incorporated into the factor A , and for our current method of taking the ratio of the output amplitude to the input amplitude, we need only worry about the *relative phase* of one to the other. This relative phase cannot be neglected in the general case, although it is often zero in important special cases.

36.3 THE TRANSFER-FUNCTION APPROACH TO INTEGRATION FORMULAS

In Chap. 21 we used a general approach to a class of formulas for computing indefinite integrals. The basic formula for integration was

$$y_{n+1} = a_0 y_n + a_1 y_{n-1} + a_2 y_{n-2} + h(b_{-1} y'_{n+1} + b_0 y'_n + b_1 y'_{n-1} + b_2 y'_{n-2})$$

where the y'_k are the integrand values and the y_k are the computed answers to the problem of computing

$$y(t) = y(0) + \int_0^t y'(x) dx$$

We propose to reexamine these formulas in the light of our present approach. Suppose, again, that the integrand (the input) is a pure sinusoid

$$y'(t) = A_I e^{2\pi i\sigma t} = A_I e^{i\omega t} \quad \omega = 2\pi\sigma$$

where the subscript I refers to input. The true answer is

$$y(t) = -\frac{iA_I}{\omega} e^{i\omega t} + C \quad C = y(0) + \frac{iA_I}{\omega}$$

Since the basic integrated formula is linear, we expect the computed values (the output) to have the same frequency, but perhaps a different phase and amplitude. Let A_0 be the output amplitude, where we allow A_0 to be complex in order to include the phase angle. Thus we can assume that the computed $y(t)$, except for roundoff, are of the form

$$y(t) = A_0 e^{i\omega t}$$

We now put these two functions into the basic formula and solve for the ratio A_0/A_I :

$$\frac{A_0}{A_I} = \frac{h(b_{-1} + b_0 e^{-i\omega h} + b_1 e^{-2i\omega h} + b_2 e^{-3i\omega h})}{1 - a_0 e^{-i\omega h} - a_1 e^{-2i\omega h} - a_2 e^{-3i\omega h}}$$

This ratio we call the *transfer function*. It is the multiplicative factor by which an input function at frequency ω is transformed to obtain the output function at the same frequency. This idea of a transfer function is a more formal statement of what we did frequently in earlier chapters when we examined what happened to a single frequency.

Using this, let us examine the special case of Simpson's formula:

$$y_{n+1} = y_{n-1} + \frac{h}{3} (y'_{n+1} + 4y'_n + y'_{n-1})$$

Here we have for our basic formula

$$\begin{array}{llll} a_0 = 0 & a_1 = 1 & a_2 = 0 & \\ b_{-1} = 1/3 & b_0 = 4/3 & b_1 = 1/3 & b_2 = 0 \end{array}$$

and the transfer function is

$$\frac{A_0}{A_I} = \frac{h(1 + 4e^{-i\omega h} + e^{-2i\omega h})}{3(1 - e^{-2i\omega h})} = \frac{-ih \cos \omega h + 2}{3 \sin \omega h}$$

which again has exactly the correct phase (the factor $-i$) for all ω and for small ωh has an amplitude like $1/\omega$.

Next, let us consider the three-eighths rule. The transfer function is

$$\frac{3h}{8} \frac{1 + 3e^{-i\omega h} + 3e^{-2i\omega h} + e^{-3i\omega h}}{1 - e^{-3i\omega h}} = (-i) \frac{3h \cos(3\omega h/2) + 3 \cos(\omega h/2)}{8 \sin(3\omega h/2)}$$

In order to standardize these formulas, we use the *ratio of the computed to the true answer* and scale ω so that $h = 1$. We have as results:

$$\text{Trapezoid rule:} \quad H_1(\omega) = \frac{\omega}{2} \cot \frac{\omega}{2}$$

$$\text{Simpson's formula:} \quad H_2(\omega) = \frac{\omega}{3} \frac{2 + \cos \omega}{\sin \omega}$$

$$\text{Three-eighths rule:} \quad H_3(\omega) = \frac{3\omega}{8} \frac{\cos(3\omega/2) + 3 \cos(\omega/2)}{\sin(3\omega/2)}$$

These three formulas are plotted in Fig. 36.3.1. We have used the $\log_{10}$ of the square of the ratio as the measure, since being twice too large is about as serious as being twice too small, and this makes logarithms the natural scale to use. We have shown these formulas plotted against both $\sigma = \omega/(2\pi)$ and the sampling rate. It is easy to see that Simpson's formula is the best of the three (for reasonably high sampling rates) as far as the size of the error is concerned. The table in Fig. 36.3.1 shows some values which are too small to read conveniently from the graph but which are important for applications.

Figure 36.3.1 also shows the curve for the formula

$$y_{n+1} = y_{n-1} + h(0.3584y'_{n+1} + 1.2832y'_n + 0.3584y'_{n-1})$$

The coefficients were determined experimentally by Leo Tick so that in the interval $0 \leq \omega \leq \pi/2$ the error would be equal-ripple (Chebyshev) and also exact for $\omega = 0$ (dc). Tick clearly used twice the Nyquist sampling rate. This formula will be discussed in more detail in Sec. 36.7.

The significance of the trapezoid rule going down while Simpson's formula goes up requires some explanation. Simpson's formula increases the amplitude of the higher frequencies whereas the trapezoid tends to smother them. Although not enough is actually known about roundoff effects, we know that sudden jumps in a function due to roundoff tend to produce some high frequencies; hence Simpson's formula amplifies these whereas the trapezoid rule tends to smother them.

If we examine Simpson's formula as a function of the sampling rate, we see that at 5 samples per cycle the error per step is about 1.5 percent, which would almost always be too big. At 7 samples per cycle we approach crude but sometimes useful accuracy, and at 10 samples per cycle we are committing an error of less than 1 part per 1,000, which in many cases is good enough.

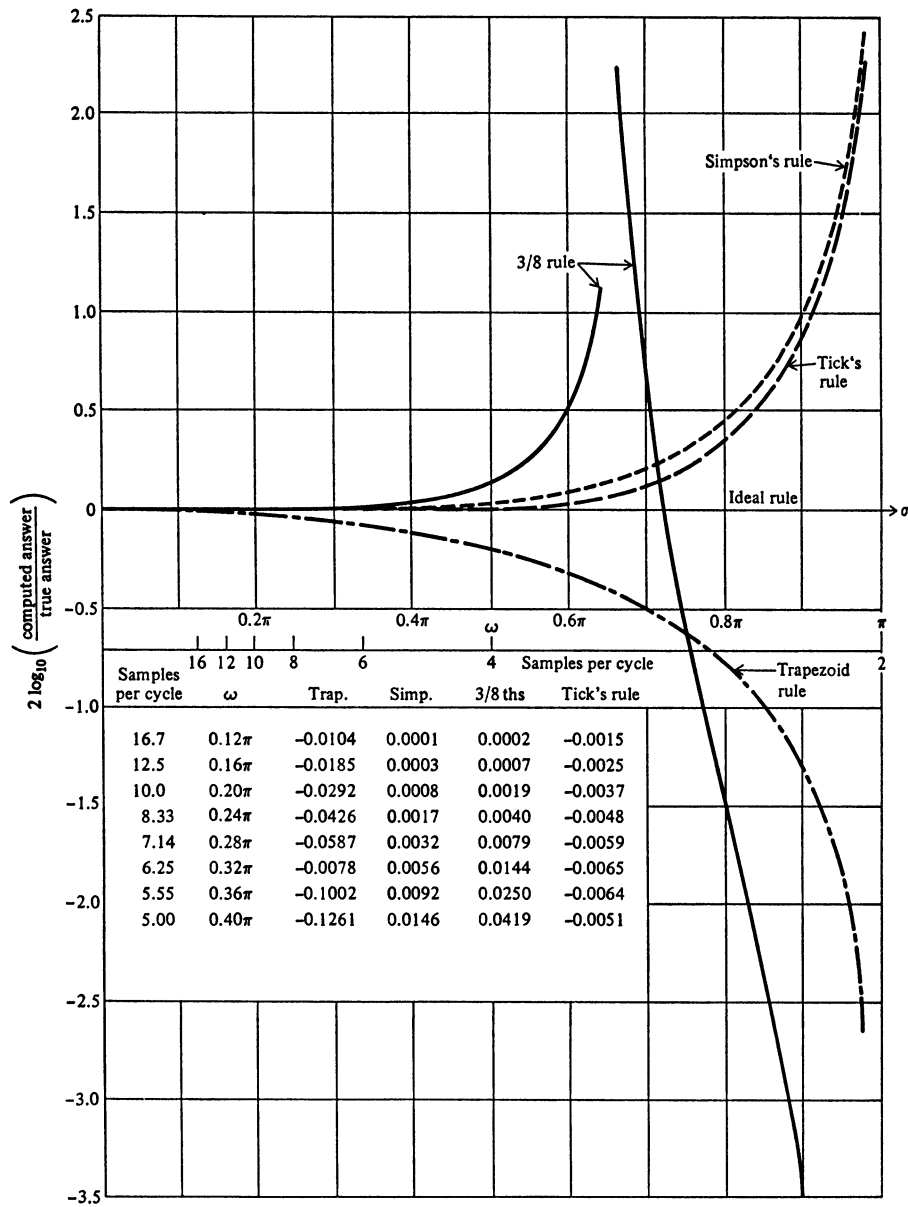


FIGURE 36.3.1
Frequency response to some integration formulas.

So far we have, in principle, been examining a single frequency $\omega = 2\pi\sigma$. Suppose that we have a general function (input)

$$f_i(t) = \int_{-\infty}^{\infty} F(\sigma) e^{2\pi i \sigma t} d\sigma$$

Each term $e^{2\pi i \sigma t}$ is transformed by the transfer function

$$G(\omega) = G(2\pi\sigma) = G_1(\sigma)$$

and we obtain for the output

$$f_o(t) = \int_{-\infty}^{\infty} F(\sigma) G_1(\sigma) e^{2\pi i \sigma t} d\sigma$$

We have on the right-hand side the transform of a product, and applying the convolution theorem (Sec. 34.7), we have

$$f_o(t) = \int_{-\infty}^{\infty} f_i(\tau) g_1(t - \tau) d\tau$$

Thus, the transform $g_1(t)$ of the transfer function $G_1(\sigma)$ convolved with the input function $f_i(t)$ gives the output function $f_o(t)$.

The transfer function $G_1(\sigma)$ contains all the information about the formula and in a sense is equivalent to the formula of integration being used.

PROBLEMS

36.3.1 Discuss the conditions necessary for an integration formula to have exactly the right phase $(-i)$.

36.4 GENERAL INTEGRATION FORMULAS

The transfer-function approach gave the exact phase for the three formulas examined in Sec. 36.3 as well as those examined earlier. When we examine the general form in the cases listed in Table 21.7 (Sec. 21.7), matters do not go so smoothly. For example, using the Adams-Bashforth method, we get (for $h = 1$)

$$\frac{9 + 19e^{-i\omega} - 5e^{-2i\omega} + e^{-3i\omega}}{24(1 - e^{-i\omega})} = -i \frac{(9e^{i\omega/2} + 19e^{-i\omega/2} - 5e^{-3i\omega/2} + e^{-5i\omega/2})}{48 \sin(\omega/2)}$$

and the numerator is not purely imaginary since the imaginary part of the parentheses is for small ω ,

$$\left(9 \sin \frac{\omega}{2} - 19 \sin \frac{\omega}{2} + 5 \sin \frac{3\omega}{2} - \sin \frac{5\omega}{2} \right) \approx \omega^5$$

The real part of the parentheses is

$$\left(9 \cos \frac{\omega}{2} + 19 \cos \frac{\omega}{2} - 5 \cos \frac{3\omega}{2} + \cos \frac{5\omega}{2} \right) \approx 24$$

so that the deviation from a purely imaginary transfer number is quite small for small ω .

This incorrect phase should not be taken too seriously. While both the amplitude and phase can be studied separately, we are concerned with the failure to have the value 1, and so we now consider

$$\left| \frac{i\omega A_0}{A_I} - 1 \right|$$

which gives a good single measure of the error.¹ The ratio we used before— $|\omega A_0/A_I|$ —can be close to 1 and still have a large phase error; this did not occur in the cases studied previously, since the phase was exactly correct.

Figure 36.4.1 shows this quantity as a function of frequency.

Table 36.4 gives the values of the quantity for the usual range of sampling used. An examination of the table shows that of the stable methods (which excludes Simpson's) the two-thirds has the least error for low and moderate sampling rates.

¹ In some problems, such as in many acoustical ones, phase errors are of minor importance; in some problems such as in the estimation of cospectra, they can be of prime importance.

Table 36.4 $\left| \frac{i\omega A_0}{A_I} - 1 \right|$ FOR VARIOUS INTEGRATION METHODS

Sampling rate	Angle	Trapezoid	Adams-Bashforth	Simpson	Three-eighths
25.00	0.08π	5.27×10^{-2}	1.05×10^{-4}	2.23×10^{-5}	5.06×10^{-5}
16.67	0.12π	1.19×10^{-2}	5.29×10^{-4}	1.14×10^{-4}	2.61×10^{-4}
12.50	0.16π	2.11×10^{-2}	1.66×10^{-3}	3.66×10^{-4}	8.49×10^{-4}
10.00	0.20π	3.31×10^{-2}	4.02×10^{-3}	9.08×10^{-4}	2.15×10^{-3}
8.33	0.24π	4.78×10^{-2}	8.25×10^{-3}	1.92×10^{-3}	4.67×10^{-3}
7.14	0.28π	6.53×10^{-2}	1.51×10^{-2}	3.66×10^{-3}	9.16×10^{-3}
6.25	0.32π	8.57×10^{-2}	2.54×10^{-2}	6.44×10^{-3}	1.68×10^{-2}

Sampling rate	Angle	One-third	One-half	Two-thirds	"S"
25.00	0.08π	5.05×10^{-5}	5.00×10^{-5}	3.44×10^{-5}	1.28×10^{-4}
16.67	0.12π	2.60×10^{-4}	2.55×10^{-4}	1.76×10^{-4}	6.24×10^{-4}
12.50	0.16π	8.39×10^{-4}	8.10×10^{-4}	5.66×10^{-4}	1.88×10^{-3}
10.00	0.20π	2.11×10^{-3}	2.00×10^{-3}	1.41×10^{-3}	4.34×10^{-3}
8.33	0.24π	4.53×10^{-3}	4.18×10^{-3}	3.02×10^{-3}	8.51×10^{-3}
7.14	0.28π	8.75×10^{-3}	7.85×10^{-3}	5.78×10^{-3}	1.49×10^{-2}
6.25	0.32π	1.57×10^{-2}	1.36×10^{-2}	1.03×10^{-2}	2.41×10^{-2}

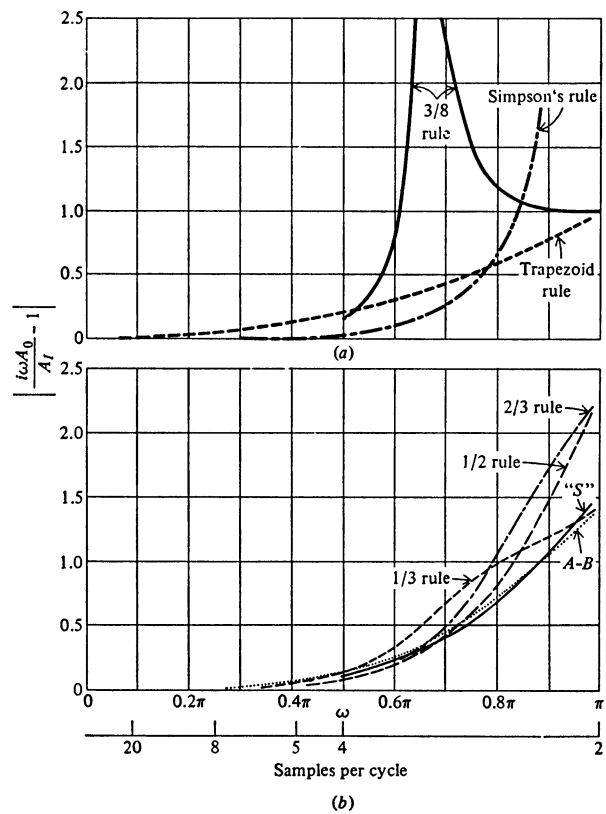


FIGURE 36.4.1

36.5 DIFFERENTIAL EQUATIONS

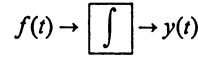
As we have observed before, there is an essential difference between computing an indefinite integral and solving an ordinary differential equation. For the purpose of clarity, suppose that the differential equation is

$$y' = Ay + f(t)$$

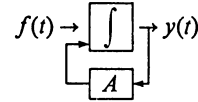
and the indefinite integral is

$$y' = f(t)$$

We have for the integral the schematic diagram



and for the differential equation



where we have a feedback loop, and need to worry even more about stability.

This difference makes a different approach more suitable, although the transfer-function approach can be used if desired. We shall use the approach of studying the size of the error in a step, rather than the ratio of output to input, since it is usually of more interest for differential equations. We are, in fact, giving different answers to the question of What accuracy? since different situations require the accuracy criterion to be applied in different ways.

We adopt the general form for integration that we have been using all along, which looks the same, as far as notation is concerned, as in Secs. 36.3 and 36.4, but is different in meaning since the y' values now come from the y values via the differential equation:

$$y_{n+1} = a_0 y_n + a_1 y_{n-1} + a_2 y_{n-2} + h(b_{-1} y'_{n+1} + b_0 y'_n + b_1 y'_{n-1} + b_2 y'_{n-2})$$

In order to study the frequency response of this formula, we substitute the function

$$y(t) = e^{2\pi i \sigma t} = e^{i\omega t} \quad \omega = 2\pi\sigma$$

and compute the error as a function of frequency.

$$\begin{aligned} G(\sigma) = G_1(\omega) &= e^{i\omega(t+h)} - a_0 e^{i\omega t} - a_1 e^{i\omega(t-h)} - a_2 e^{i\omega(t-2h)} \\ &\quad - h i \omega (b_{-1} e^{i\omega(t+h)} + b_0 e^{i\omega t} + b_1 e^{i\omega(t-h)} + b_2 e^{i\omega(t-2h)}) \\ |G_1(\omega)| &= |1 - a_0 e^{-i\omega h} - a_1 e^{-2i\omega h} - a_2 e^{-3i\omega h} \\ &\quad - i h \omega (b_{-1} + b_0 e^{-i\omega h} + b_1 e^{-2i\omega h} + b_2 e^{-3i\omega h})| \end{aligned}$$

If now we use a general function

$$y(t) = f(t) = \int_{-\infty}^{\infty} F(\sigma) e^{2\pi i \sigma t} d\sigma$$

we find that the error is

$$R[f(t)] = \int_{-\infty}^{\infty} F(\sigma) G(\sigma) e^{2\pi i \sigma t} d\sigma$$

Using the convolution theorem, we get

$$R[f(t)] = \int_{-\infty}^{\infty} f(s)g(t-s) ds = \int_{-\infty}^{\infty} g(t-s)y(s) ds$$

In this form the error may be compared with the error form of Chap. 16. In particular, we had

$$R[f(t)] = \int_A^B f^{(m)}(s)G(s) ds$$

where $G(s)$ was the influence function and was composed of terms of the form

$$(x_k - s)_+^l$$

We again see a significant difference between the two theories: The polynomial approach leads us to express the error as an integral of the m th derivative of the function times an influence function, whereas the Fourier-integral approach leads us to use *the function itself* or, alternatively, its transform rather than the m th derivative and of course a different influence function. However, this last form is deceptive; if the function $g(t-s)$ is examined closely, then it is seen to be merely the original formula in disguise.

36.6 CHEBYSHEV DESIGN OF INTEGRATION FORMULAS: THEORY

The availability of a method of evaluation leads sooner or later to a new method of design. So many different possibilities arise in practice that we cannot hope to cover them all; rather we shall content ourselves with sketching an approach to Chebyshev design, as this type is becoming increasingly important. One of the aims of this book has been to gradually bring the reader to the position where he is able to use the principles given in the text to design his own formulas, and we believe that this point has now been reached.

Suppose, then, that we wish to design a method for integrating differential equations such that the error in the frequency domain has a Chebyshev form rather than the form that we have had so far—very good at the origin and increasingly poor as the frequency increases. For some purposes the Chebyshev error form is clearly preferable to the power-series error form.

We begin by transforming our range of frequencies in ω so that they lie between -1 and $+1$; hence we can now use the standard Chebyshev polynomials. If we adopt the general form of the previous section, we are led to the G function for the error at frequency ω

$$G(\sigma) = e^{t\omega(t+h)} - a_0 e^{t\omega t} - a_1 e^{t\omega(t-h)} - a_2 e^{t\omega(t-2h)} \\ - ih(b_{-1} e^{t\omega(t+h)} + b_0 e^{t\omega t} + b_1 e^{t\omega(t-h)} + b_2 e^{t\omega(t-2h)})$$

We now recall the observation that if we wish to have the error in the form of a Chebyshev polynomial, then we should consider getting the whole expression in Chebyshev polynomials. For this we will need to express $e^{i\omega z}$ in terms of Chebyshev polynomials. A search of Watson's "Bessel Functions" [62] reveals the generating function

$$e^{(z/2)(t-1/t)} = \sum_{n=-\infty}^{\infty} t^n J_n(z)$$

for the Bessel functions $J_n(z)$. Set $t = ie^{i\theta}$ and use the obvious relation (from the generating function)

$$J_{-n}(z) = (-1)^n J_n(z)$$

which is the same as

$$i^{-n} J_{-n}(z) = i^n J_n(z)$$

to get

$$e^{iz \cos \theta} = J_0(z) + 2 \sum_{n=1}^{\infty} i^n J_n(z) \cos n\theta$$

We set $\cos \theta = \omega$, and we have the desired expansion of $e^{i\omega z}$ in terms of Chebyshev polynomials

$$e^{iz\omega} = J_0(z) + 2 \sum_{n=1}^{\infty} i^n J_n(z) T_n(\omega) \quad -1 \leq \omega \leq 1$$

In $G(\sigma)$ we factor out the term $e^{i\omega t}$, and we have

$$|G(\sigma)| = |e^{i\omega h} - a_0 - a_1 e^{-i\omega h} - a_2 e^{-2i\omega h} - i h \omega (b_{-1} e^{i\omega h} + b_0 + b_1 e^{-i\omega h} + b_2 e^{-2i\omega h})|$$

Using the expansion of the exponential with $z = h, 0, -h, -2h$, in turn, and the appropriate identities involving $T_k(\omega)$, we can arrange everything in an expansion in Chebyshev polynomials with complex coefficients (involving Bessel functions).

If we now start equating coefficients to zero, we get one equation for each complex coefficient, and we could proceed as far as we wished, but we would have to save a parameter or two for stability and other desirable properties. The result would be an error curve that did not pass through zero at zero (dc) frequency, and this would probably, though not always, cause us some embarrassment; the simple equation

$$y' = 0 \quad y(0) = A \neq 0$$

would have a solution that was not a constant.

To remedy this defect, we start by requiring that zero frequency fit exactly, that is, require that

$$1 = a_0 + a_1 + a_2$$

We then begin equating coefficients to zero. Thus we are sure that (within round-off) dc is computed correctly. Note that if we keep the same frequency band and change the sampling rate by changing h , then *all* the coefficients change, rather than a factor h merely changing as in the polynomial case.

Thus we see the main outlines of the Chebyshev approach. We see how to find the desired formula, provided we are willing to do some work to obtain it. And this is true over a wide range of formulas—the general approach frequently works provided we do some hard work to obtain it, and at times use a little imagination to overcome a few messy spots. The art of finding formulas has been practically reduced to a science; the methods developed so far suffice to provide more formulas than there is room to list in a book many times thicker than this one, since there are many combinations of ideas that can be used in different circumstances. It is hoped that the reader will now feel that he can, with some labor, design formulas to fit the situation rather than force the situation to fit the classical formulas.

36.7 SOME DETAILS OF CHEBYSHEV DESIGN

Since the Chebyshev design methods are not readily available in the literature, we indicate some of the details for integration formulas. Let the integration formula be of the standard form

$$y_{n+1} = a_0 y_n + a_1 y_{n-1} + a_2 y_{n-2} + h(b_{-1} y'_{n+1} + b_0 y'_n + b_1 y'_{n-1} + b_2 y'_{n-2})$$

We assume, as in the Fourier approach, that

$$y(t) = e^{i\omega t}$$

and using the Chebyshev expansion of $e^{i\omega t}$, we have

$$y(t) = J_0(t) + 2 \sum_{n=1}^{\infty} i^n J_n(t) T_n(\omega)$$

The derivative is

$$y'(t) = J'_0(t) + 2 \sum_{n=1}^{\infty} i^n J'_n(t) T_n(\omega)$$

We now substitute these two expressions into our basic integration formula, using the appropriate values of t , namely, $t = h, 0, -h, -2h$, and factor out the term $e^{i\omega h}$. We get immediately

$$\begin{aligned}
 J_0(h) + 2 \sum_{n=1}^{\infty} i^n J_n(h) T_n(\omega) &= a_0 J_0(0) \\
 &+ a_1 \left[J_0(-h) + 2 \sum_{n=1}^{\infty} i^n J_n(-h) T_n(\omega) \right] \\
 &+ a_2 \left[J_0(-2h) + 2 \sum_{n=1}^{\infty} i^n J_n(-2h) T_n(\omega) \right] \\
 &+ b_{-1} h \left[J'_0(h) + 2 \sum_{n=1}^{\infty} i^n J'_n(h) T_n(\omega) \right] \\
 &+ b_0 h [2i J'_1(0) T_1(\omega)] \\
 &+ b_1 h \left[J'_0(-h) + 2 \sum_{n=1}^{\infty} i^n J'_n(-h) T_n(\omega) \right] \\
 &+ b_2 h \left[J'_0(-2h) + 2 \sum_{n=1}^{\infty} i^n J'_n(-2h) T_n(\omega) \right]
 \end{aligned}$$

We now rearrange this to have the form of an expansion in Chebyshev polynomials, and we equate coefficients of T_n [since the $T_n(\omega)$ are linearly independent]

$T_0(\omega)$:

$$J_0(h) = a_0 + a_1 J_0(h) + a_2 J_0(2h) + h[b_{-1} J'_0(h) - b_1 J'_0(h) - b_2 J'_0(2h)]$$

$$\frac{T_1(\omega)}{2i}:$$

$$J_1(h) = -a_1 J_1(h) - a_2 J_1(2h) + h[b_{-1} J'_1(h) + b_0 J'_1(0) + b_1 J'_1(h) + b_2 J'_1(2h)]$$

$$\frac{T_k(\omega)}{2(i)^k}:$$

$$\begin{aligned}
 J_k(h) &= (-1)^k [a_1 J_k(h) + a_2 J_k(2h)] + h[b_{-1} J'_k(h) - (-1)^k b_1 J'_k(h)] \\
 &\quad - (-1)^k b_2 J'_k(2h)
 \end{aligned}$$

As a first (and degenerate) case, consider the equivalent of the trapezoid rule; that is, set $a_1 = a_2 = b_1 = b_2 = 0$. We shall also require that the formula be exact for $y \equiv 1$ (the dc case), which amounts to setting

$$a_0 = 1$$

We now take the coefficients of $T_0(\omega)$ and $T_1(\omega)$

$$J_0(h) = 1 + hb_{-1} J'_0(h)$$

$$J_1(h) = h[b_{-1} J'_1(h) + b_0 J'_1(0)]$$

From these we get [where $J'_0(h) = -J_1(h)$]

$$b_{-1} = \frac{1 - J_0(h)}{hJ_1(h)}$$

$$b_0 = 2 \frac{J_1(h) - 2hb_{-1}J'_1(h)}{h}$$

as the coefficients of the integration formula.

In order to understand (and check) these equations, we examine what happens as h approaches zero. We know [62] that

$$J_n(z) = \frac{1}{n!} \left(\frac{z}{2}\right)^n \left[1 - \frac{(z/2)^2}{n+1} + \frac{(z/2)^4}{2!(n+1)(n+2)} - \frac{(z/2)^6}{3!(n+1)(n+2)(n+3)} + \cdots \right]$$

In particular

$$J_0(h) = 1 - \frac{h^2}{4} + \frac{h^4}{64} - \cdots$$

$$J_1(h) = \frac{h}{2} - \frac{h^3}{16} + \frac{h^5}{384} - \cdots$$

$$J_2(h) = \frac{h^2}{8} - \frac{h^4}{96} + \frac{h^6}{3,072} - \cdots$$

Hence as $h \rightarrow 0$,

$$b_{-1} = \frac{h^2/4 - h^4/64 + \cdots}{h^2/2 - h^4/16 + \cdots} \simeq \frac{1 - h^2/16}{2(1 - h^2/8)} \rightarrow \frac{1}{2}$$

$$b_0 = \frac{h - h^3/8 + \cdots - 2b_{-1}h(\frac{1}{2} - 3h^2/16 + \cdots)}{h} \rightarrow \frac{1}{2}$$

At first it may seem surprising that the two coefficients b_{-1} and b_0 are not equal, but a little thought shows that there was nothing else to be expected. The difficulty lies in the fact that we made the error curve go through zero at $\omega = 0$ and then tried to make the dominant term in the error expansion proportional to $T_2(\omega)$, which clearly does not have a zero at $\omega = 0$. The case is really too trivial to be meaningful.

The next case uses two old values of the function and the derivative, setting only $a_2 = b_2 = 0$. In this case we save one parameter for stability. Thus we have the equations

$$\begin{aligned} \text{dc:} \quad & 1 = a_0 + a_1 \\ T_0(\omega): \quad & J_0(h) = a_0 + a_1 J_0(h) + h[b_{-1}J'_0(h) - b_1J'_0(h)] \\ T_1(\omega): \quad & J_1(h) = -a_1 J_1(h) + h[b_{-1}J'_1(h) + b_0J'_1(0) + b_1J'_1(h)] \\ T_2(\omega): \quad & J_2(h) = a_1 J_2(h) + h[b_{-1}J'_2(h) - b_1J'_2(h)] \end{aligned}$$